

Photovoltaics Report

Fraunhofer Institute for Solar Energy Systems ISE
with the support of PSE Projects GmbH

Freiburg, 14 July 2026
www.ise.fraunhofer.de

CONTENT

- Quick Facts
- Topics:
 - PV Market
 - Solar Cells / Modules / System Efficiency
 - Life cycle assessment (LCA) and sustainability aspects
 - Price Development
- Abbreviations
- Further Studies and Analyses
- Acknowledgements

Introduction

Preliminary Remarks

- The intention of this presentation is to provide up-to-date information. However, facts and figures change rapidly, and the given information may soon be outdated again.
- This work has been carried out under the responsibility of Dr. Simon Philipps (Fraunhofer ISE) and Werner Warmuth (PSE Projects GmbH).
- Price indications are always to be understood as nominal, unless stated explicitly. For example, prices in the learning curves are inflation adjusted.
- The slides have been made as accurate as possible. Please send any comments or suggestions for improvement to both simon.philipps@ise.fraunhofer.de and warmuth@pse-projects.de
- Please cite the information presented in these slides as follows:
©Fraunhofer ISE: Photovoltaics Report, updated: 14 July 2026

Quick Facts

Parameter	Value	Status	Reference	Date of data
Germany / EU27 / Worldwide				
PV installation market [GW]	17.5 / 65.6 / 602 GW 17.2 / 65.1 / 698 GW	End of 2024 End of 2025	BNA / SPE / IEA BNA / SPE / IEA	06/2026; 01/2026; 04/2025 06/2026; 01/2026; 04/2026
Cumulative PV installation [GW]	92.5 _{net} / 265.8 _{net} / 2,276 GW 107.6 _{net} / 309.9 _{net} / 2,974 GW	End of 2024 End of 2025	ISE / ISE / IEA ISE / ISE / IEA	05/2026; 05/2026; 04/2026 05/2026; 05/2026; 04/2026
PV power generation [TWh]	59.7 _{net} / 308.4 _{gross} / 2166.9 _{gross} 70,1 _{net} / 372.1 _{gross} / 2811.1 _{gross}	2024 2025	ISE / EI / EI ISE / EI / EI	07/2026; 07/2026; 07/2026 07/2026; 07/2026; 07/2026
PV electricity share	14.5% _{net} / 11.0% _{gross} / 6.9% _{gross} 16.9% _{net} / 13.0% _{gross} / 8.7% _{gross}	2024 2025	ISE / EI / EI ISE / EI / EI	07/2026; 07/2026; 07/2026 07/2026; 07/2026; 07/2026
Worldwide				
c-Si share of production	98%	2024	ITRPV	03/2025
Record solar cell efficiency: III-V MJ (conc.) /mono-Si /Perovskite /Pero-Si / CdTe	47.6 / 28.1 / 26.9 / 35.2 / 21.0	2025	Green et al. Version 68	05/2026
Germany				
Price PV rooftop system (10 kWp)	900 to 1,500 €/kWp	End of 2025	gruenes.haus, other	03/2026
LCOE PV power plant	4.1 to 5.0 ct€/ kWh	2024	ISE	08/2024
Lowest/Latest PV-tender price (average, volume-weighted value)	4.33 ct€/ kWh 4.66 ct€/ kWh	02/2018; 03/2025	BNA	04/2025

Executive Summary

PV Market: Global

- Photovoltaics is a fast-growing market: The Compound Annual Growth Rate (CAGR) of cumulative PV installations was about 27% between the years 2015 and 2025.
- Wafer size increased. Keeping the same number of cells, larger PV module sizes are realized, allowing a power range of up to 750 W per module.
- In 2025, Europe's contribution to the total cumulative PV installations amounted to 17%. In contrast, installations in China accounted for 51% (48% in 2024) and in North America for 9% respectively.
- Silicon wafer-based technology accounted for about 98% of total production in 2025 with a 86% share of n-type wafers according to ITRPV. Monocrystalline n-type TOPCon technology has become dominant.
- Market shifts from subsidy-driven to a competitive pricing model (Power Purchase Agreements PPA).
- In addition to building-integrated systems (on roofs or building facades) and ground-mounted systems, more and more PV systems are being installed on agricultural land (agrivoltaics) and on bodies of water (floating PV). Furthermore, vehicle-integrated PV is entering the market.
- With increasing share of power generated by renewables, the integration of batteries with energy management systems is becoming increasingly important.

Executive Summary

PV Market: Focus Germany

- In 2025, PV installations in Germany (approximately 5.7 million PV systems with a capacity of over 800 W) accounted for about 4% (108 GWp) of the cumulative PV capacity installed worldwide (2974 GWp). According to the BNA, newly installed capacity in Germany amounted to approximately 16.6 GWp in 2025, compared to 16.9 GWp in 2024.
- In 2025, PV accounted for 16.9% of net electricity generation and all renewable energies for around 62%.
- In 2025 GHG emissions of about 63 million tons CO₂ equivalents were avoided due to 87 TWh PV electricity consumed in Germany. Of this, around 71 TWh was fed into the public grid, while a remarkable 16.9 TWh was used by the PV system operators themselves for their own power needs.
- Solar and wind energy have the lowest levelized cost of electricity.
- Nowadays, solar power systems for private homes and small businesses are often installed alongside battery storage systems, heat pumps, and electric vehicle charging stations. Due to relatively high electricity tariffs in Germany, self consumption is the prevailing business model. The installation of balcony solar systems is another growing trend.
- With increasing generation capacity from solar and wind, the flexible integration of volatile electricity into the grid becomes more important. Grid expansion, load management, smart grids, bidirectional charging of vehicle batteries, etc. must be promoted in order to avoid curtailment of renewable power plants.

Executive Summary

Solar Cell / Module Efficiencies

- The record lab cell efficiency* is 28.1% for monocrystalline wafer-based technology. The highest lab efficiency in thin film technology is 23.4% for CIGS and 21.0% for CdTe solar cells. Record lab cell efficiency for perovskite solar cells is 26.9% and for Pero-Si 35.2%.
- Over the past decade, the efficiency of commercial silicon modules based on monocrystalline wafers has risen from about 17% to just under 25%. At the same time, the CdTe module efficiency increased from 15.6% to almost 20%.
- In the laboratory, the best performing modules are based on monocrystalline silicon with 26.0% efficiency. Record efficiencies demonstrate the potential for further efficiency increases at the production level.
- In the laboratory, high concentration multi-junction solar cells achieve an efficiency of up to 47.6% today. With concentrator technology, module efficiencies of up to 38.9% have been reached.

Executive Summary

Energy Payback Time

- Silicon usage for silicon cells has been reduced significantly during the last 20 years from around 16 g/Wp (in 2004) to about 2.0 g/Wp in 2024 due to increased efficiencies, thinner wafers (140 μm) using diamond wire saws, and larger ingots.
- The Energy Payback Time of PV systems is dependent on the geographical location: PV systems manufactured in Europe and installed in Northern Europe require approximately 1.1 years to pay back the energy input, while PV systems installed in the South require 0.9 years to pay back the energy input, depending on the technology installed and the grid efficiency.
- A PV system located in Sicily using wafer-based silicon modules has an Energy Payback Time of about one year. Assuming a 20-year lifetime, this type of system can produce twenty times the energy invested in it.
- PV modules can be recycled, recovering rare and valuable materials. Further research and development is needed to make these recycling processes more in-depth and cost-effective.

Executive Summary

Price Development

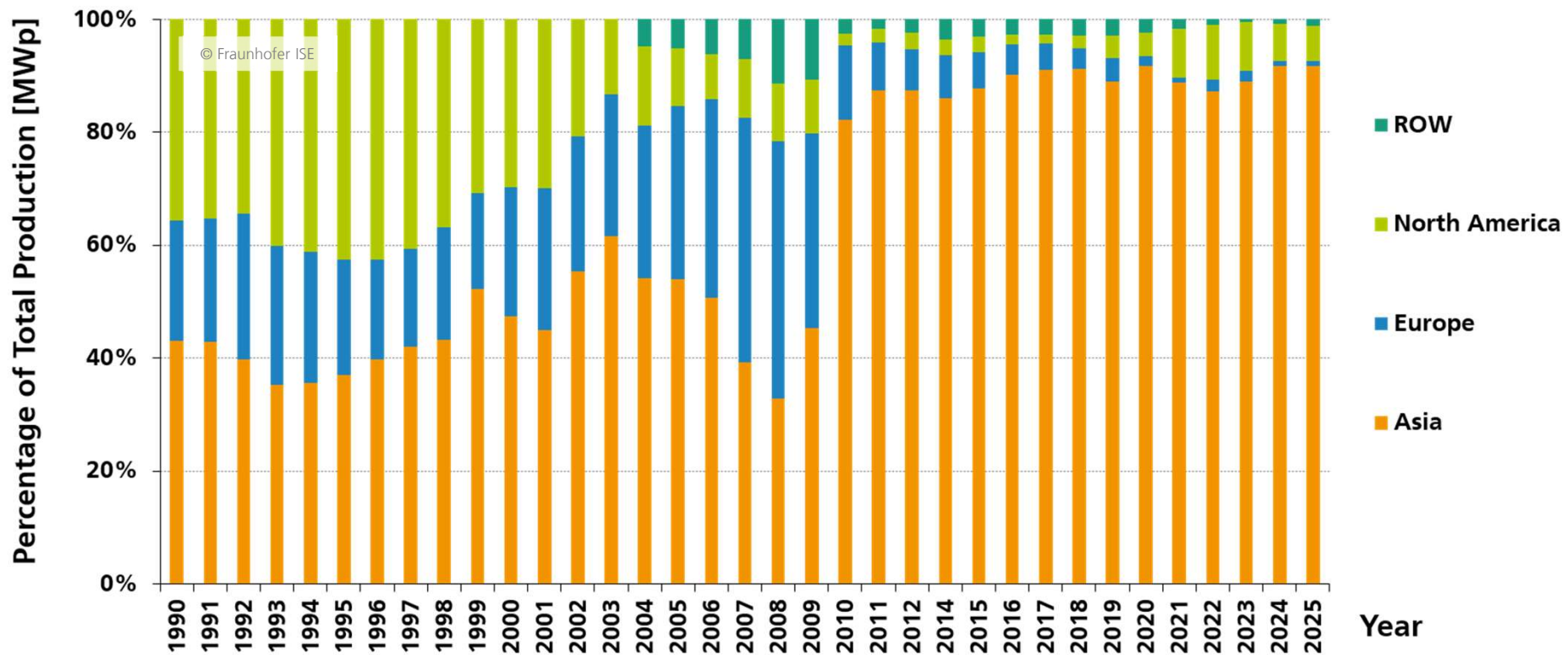
- PV module prices remain near record lows despite upstream cost pressures, with future pricing hinging on Chinese demand, overcapacity, and policy-driven consolidation. China's April 2026 removal of the 9% export VAT refund, together with rising material and shipping costs, is increasing global solar module and installation prices.
- In Germany, a typical 10 to 100 kWp PV rooftop-system cost around 14,000 €/kWp in 1990. At the end of 2025, such systems cost less than 8% of that in 1990. Over the last 35 years, the compound annual growth rate (CAGR) of net prices was about -7%. The total installed costs of utility-scale PV systems in Germany have risen by around 40% compared to the previous year.
- The Experience Curve (also called Learning Curve) shows that in the last 45 years the module price decreased by 26.7% with each doubling of the cumulated global module production. Cost reductions result from economies of scale and technological improvements. The global average selling price (ASP) was about 0.10 US\$/Wp in 2025.

1. PV Market

- By region
- By technology

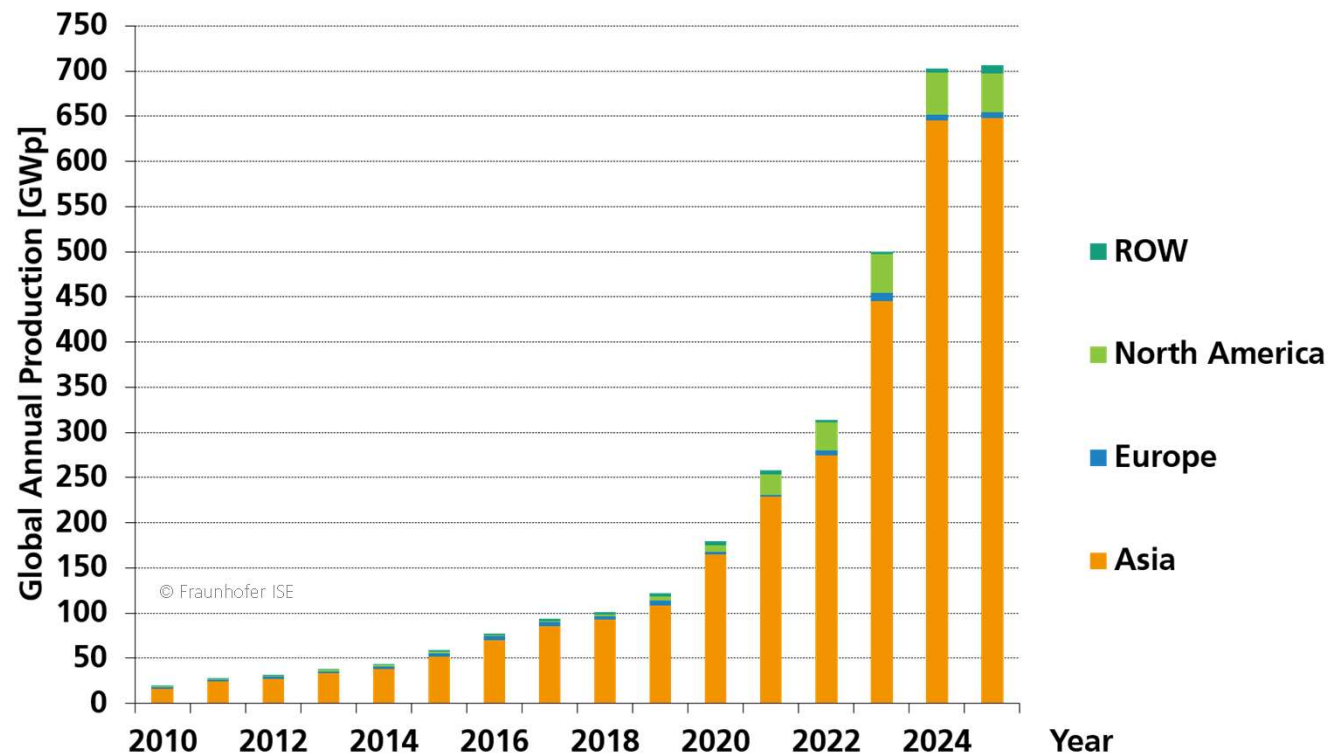
PV Module Production by Region 1990-2025

Percentage of Total MWp Produced



PV Module Production by Region

Global Annual Production

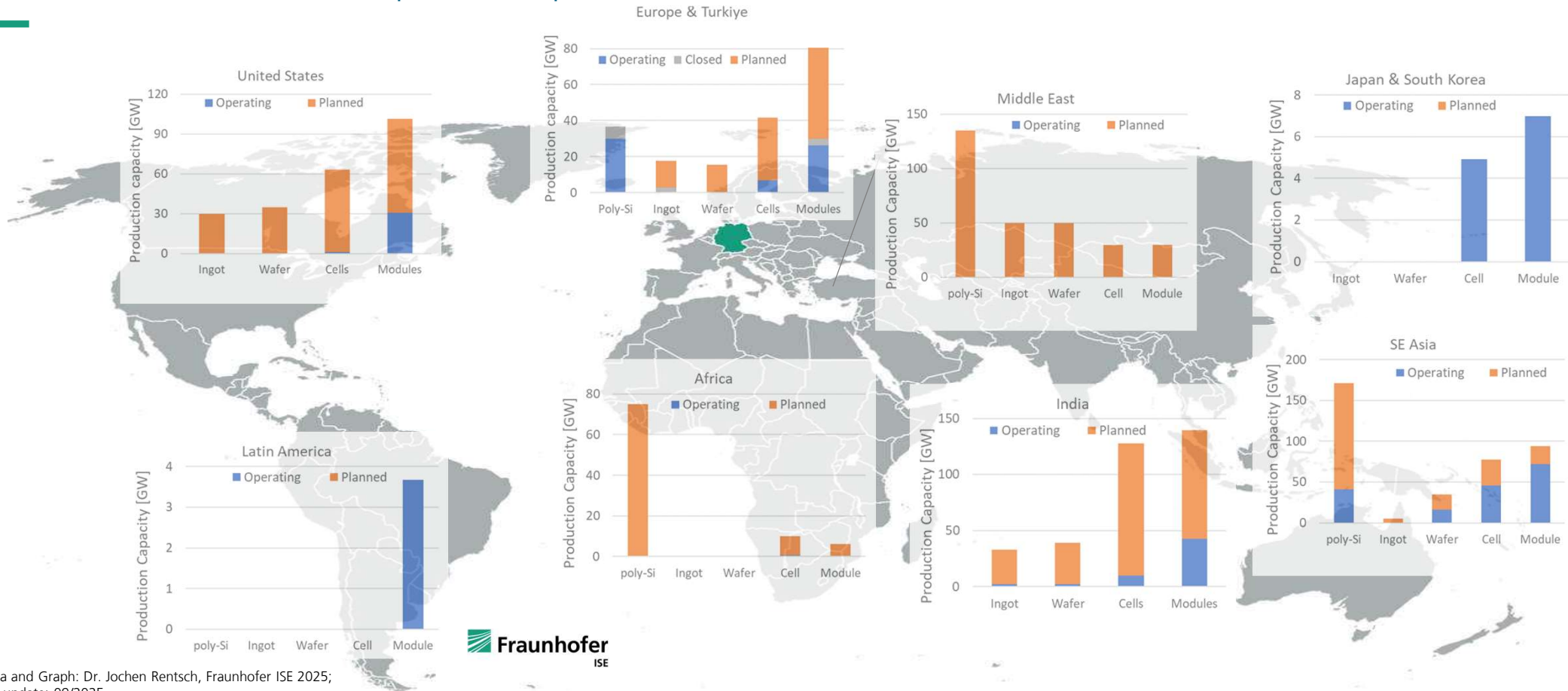


Annual production has increased about 12-fold over the past decade. In 2025, approximately 96% of solar modules and their components came from Asia. Global PV module shipments stagnated from 2024 to 2025 due to a market correction following unsustainably high shipment volumes in 2024. This was compounded by significant policy and tariff shifts in major markets such as the US and Europe, as well as an industry-wide shift towards prioritising profitability over volume.

Data from 2000 to 2009: Navigant; from 2010 to 2021 IHS Markit; from 2022 estimates based on IEA and other sources. Graph: PSE Projects GmbH 2026. Date of data 07/2026 – preliminary estimates

Global Production Capacities (w/o China)

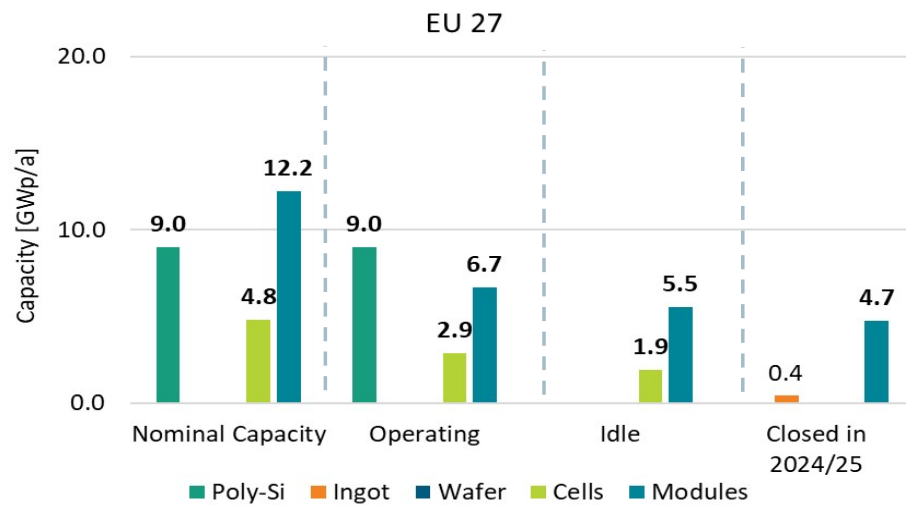
Current and Announced Capacities September 2025



Data and Graph: Dr. Jochen Rentsch, Fraunhofer ISE 2025;
last update: 09/2025

EU PV Manufacturing Landscape – Status Quo

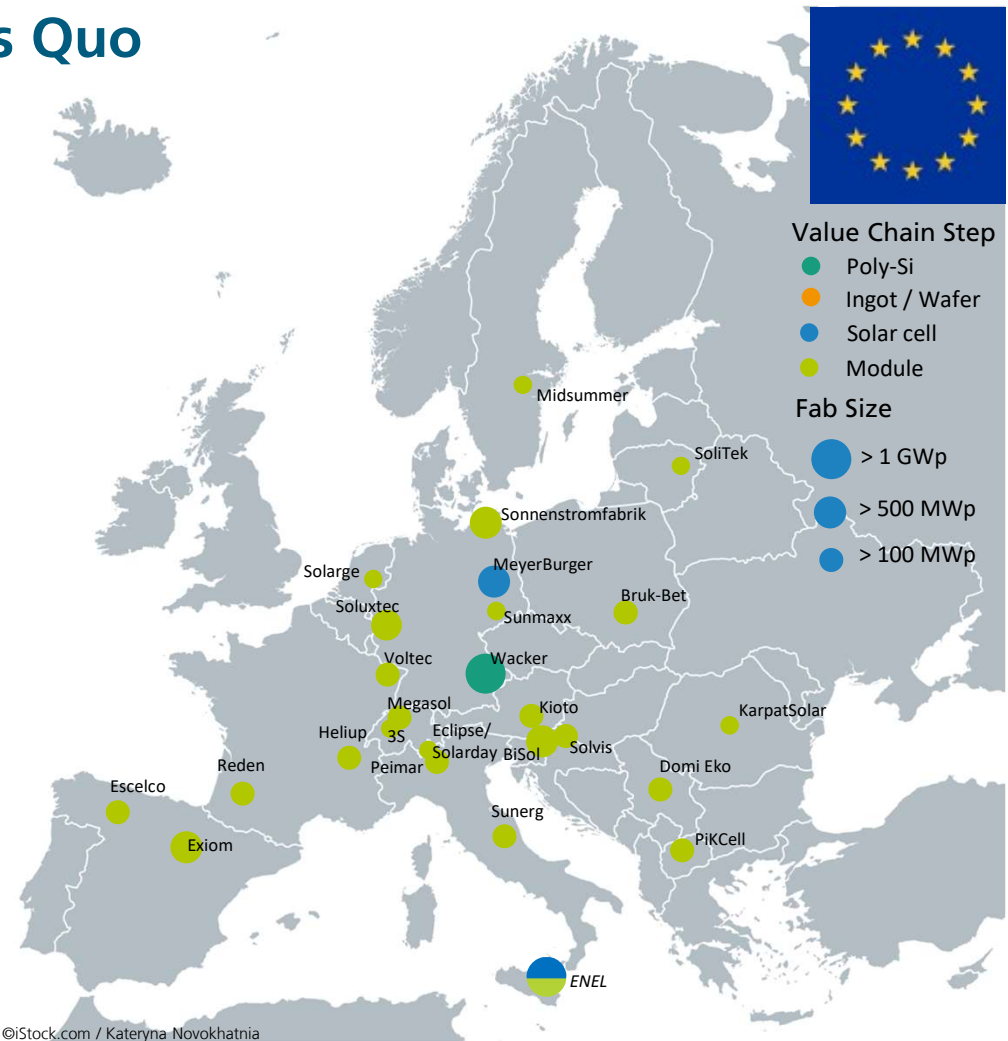
EU 27



Status Quo in the EU

- Many PV Module manufacturers still exist. But almost all of them with a small production capacity (< 1 GWp/a).
- Cell production capacity of <5 GWp/a.
- No active Ingot & Wafer production.
- One Polysilicon manufacturer.

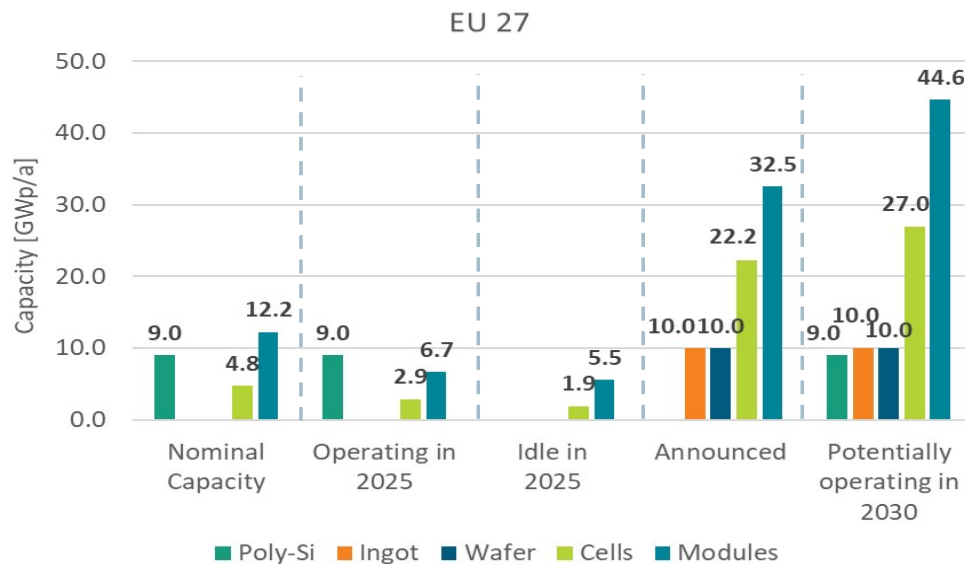
Data and Graph: Dr. Jochen Rentsch, Fraunhofer ISE 2025; last update: 08/2025



Source: ©iStock.com / Kateryna Novokhatnia

EU PV Manufacturing Landscape – Perspective

Announced / Planned GW Fabs

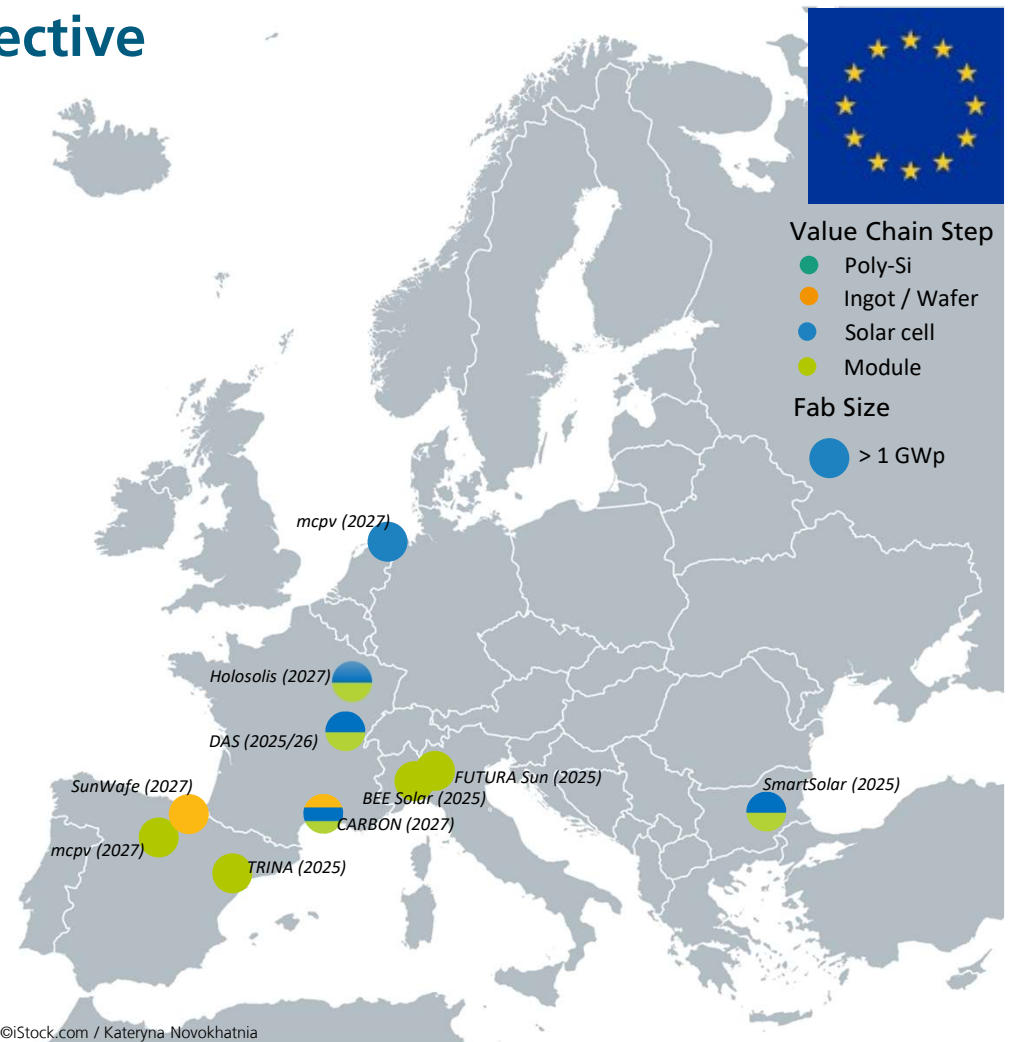


Diversification of European PV supply:

- GWp PV Manufacturing projects in Europe announced.
- What is planned in other global regions?

Data and Graph: Dr. Jochen Rentsch, Fraunhofer ISE 2025; last update: 08/2025

Source: ©iStock.com / Kateryna Novokhatnia



PV Production in Germany - Status Quo

PV Module Suppliers – November 2024

Company	Location	Capacity [MW]	Website
Soluxtec	Bitburg	1200	https://www.soluxtec.de/
Sonnenstromfabrik	Wismar	525	https://www.sonnenstromfabrik.com/de/
Solarwelt (Heckert)	Langenwetzendorf	400	https://www.heckertsolar.com/standort-lwd/
Heckert Solar	Chemnitz	400	https://www.heckertsolar.com
Aleo Solar	Berlin	300	https://www.aleo-solar.de/
Heliatek	Dresden	250	https://www.heliatek.com/de/
OPES	Zwenkau	200	https://www.opes-solutions.com/de/
Avancis	Torgau	100	https://www.avancis.de/
AxSun	Laupheim	50	https://www.axsun.de/
Sunmaxx	Ottendorf-Okrilla	50	https://sunmaxx-pvt.com/de
Antec Solar	Arnstadt	50	https://www.antec.solar/

Data and Graph: Jochen Rentsch, Fraunhofer ISE 2025; last update: 11/2024

Value-added stage

- mg-Si
- Poly-Si
- Ingot / Wafer
- Solar cell
- Module

Factory size

- > 1 GWp
- > 500 MWp
- > 100 MWp
- > 50 MWp

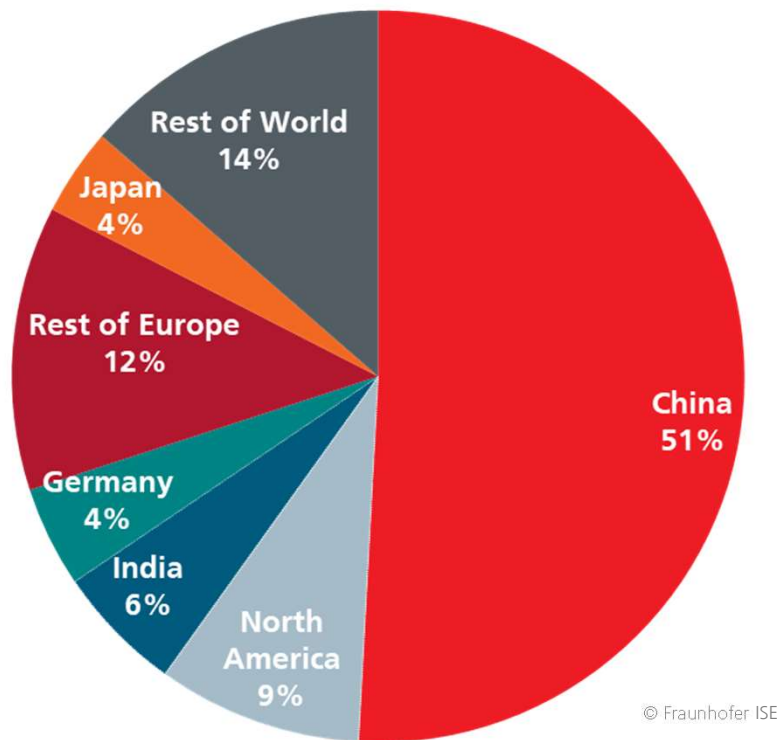


Status 11/24

Source: ©iStock.com / Kateryna Novokhatnia

Global Cumulative PV Installation by Region

Status 2025



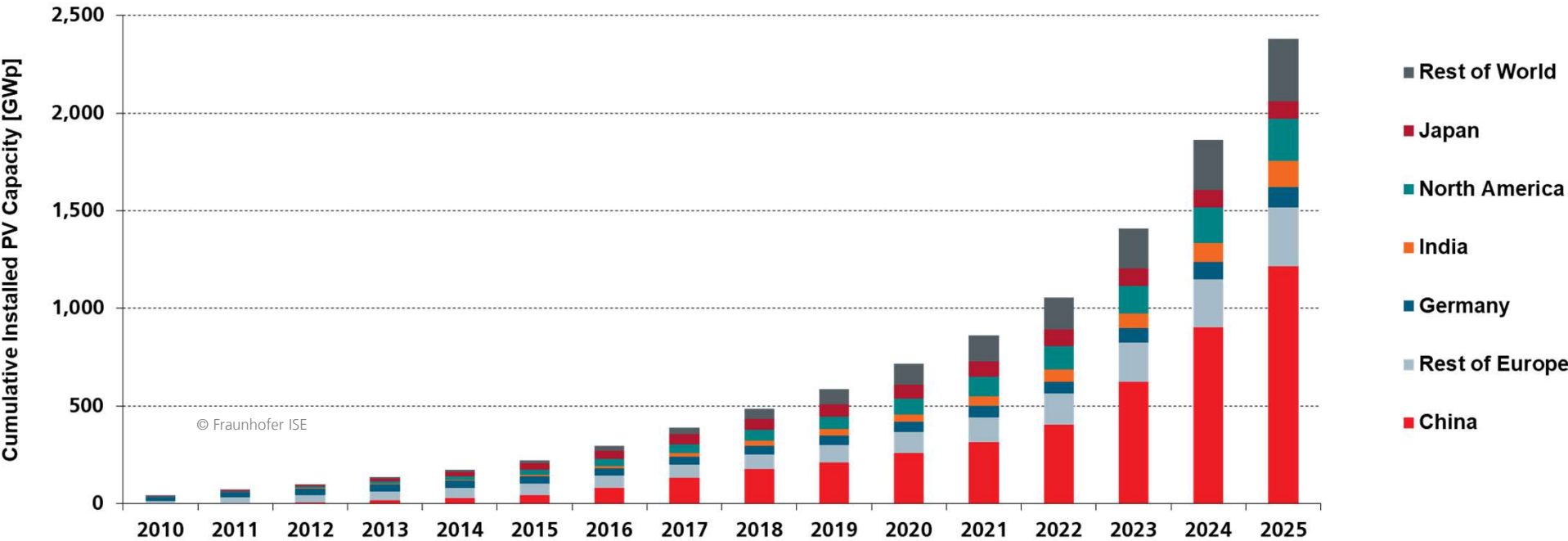
According to IEA-PVPS, the total cumulative installations amounted to approximately 2,973 GWp at the end of 2025. IRENA reports 2,381.2 GWp (on-grid) and 10.3 GWp (off-grid), totalling 2,391.6 GWp for the same year.

All percentages relate to global installed PV capacity, including off-grid systems.

Source: IEA-PVPS Snapshot of Global Market; Graph data: IRENA 2026. Graph: PSE Projects GmbH 2026. Date of data: 04/2026

Global Cumulative PV Installation

By Region

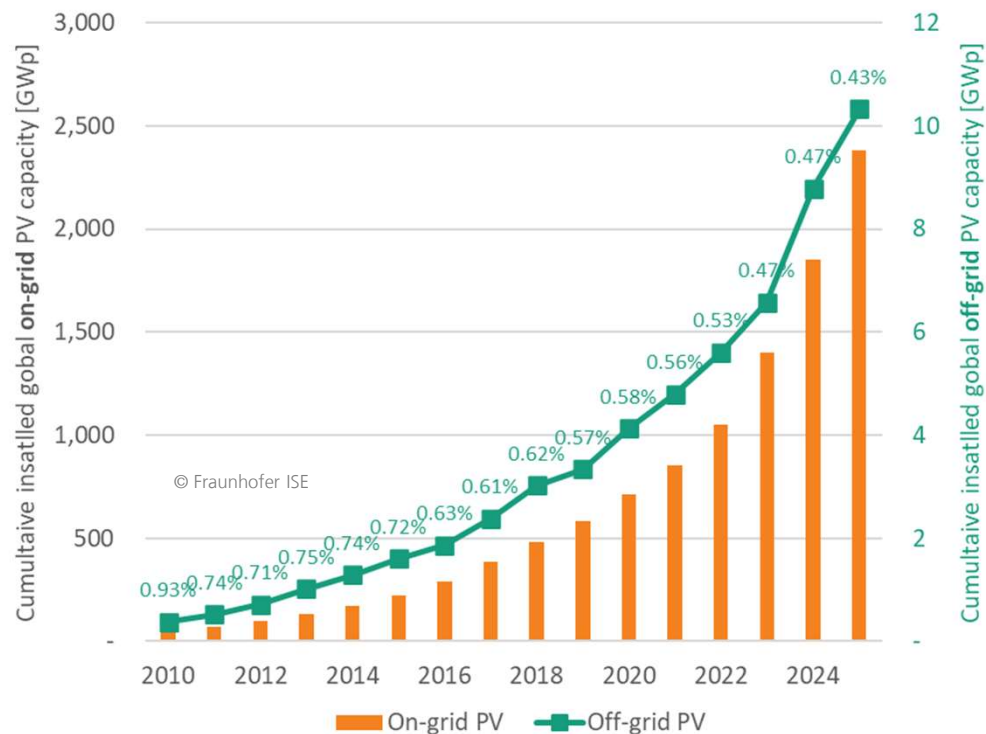


Note: Cumulative installed capacity is considered here as 'aggregate annual additions.'

Data: IRENA 2026. Graph: PSE Projects GmbH 2026. Date of data: 04/2026

Global Cumulative PV Installation

Cumulative Installed On-Grid and Off-Grid Capacity (2010-2025)



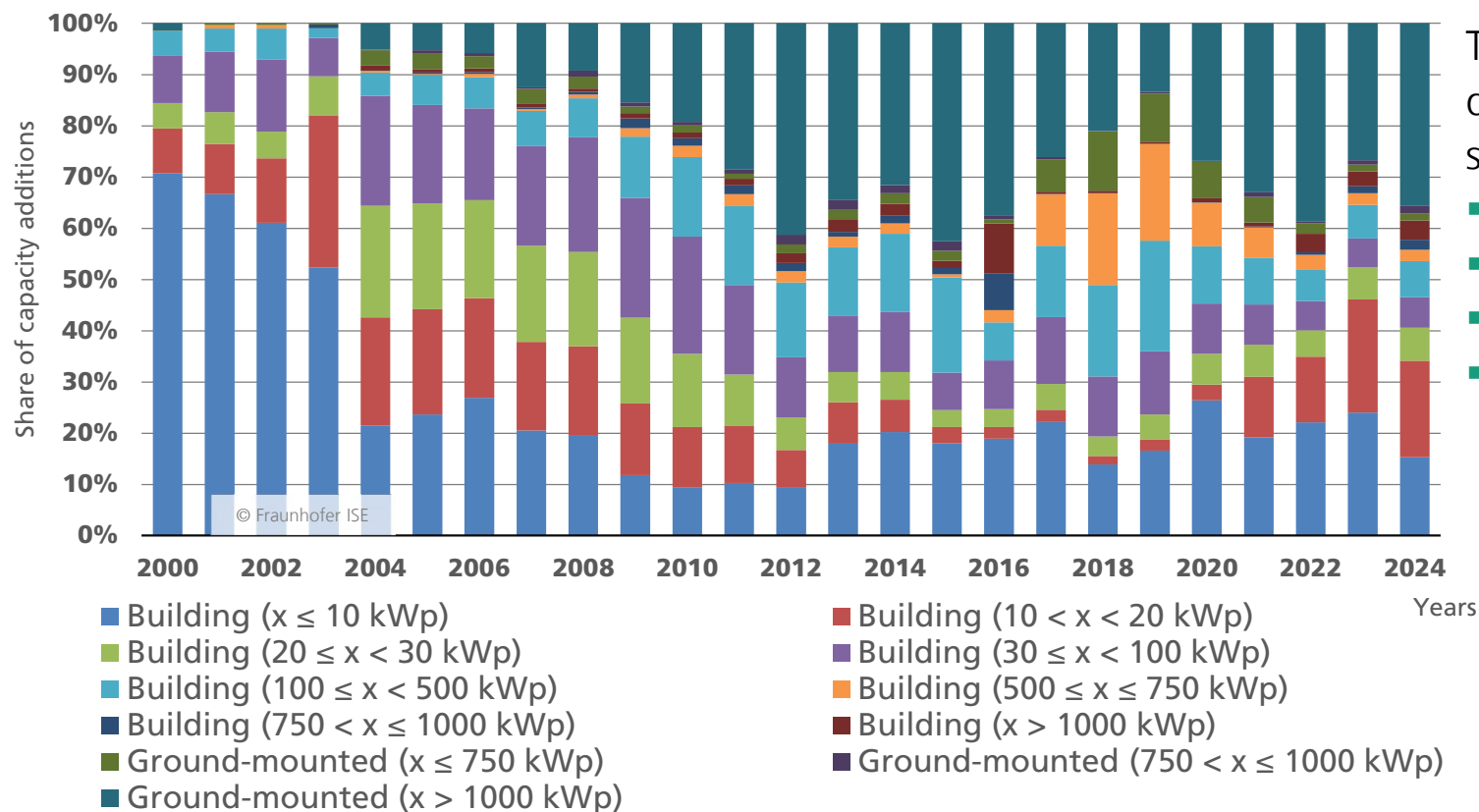
Percentages show the share of cumulative off-grid PV installations in relation to total cumulative PV installations. Approximately 99.6% of today's installed PV capacity is connected to the grid.

The proportion of off-grid systems compared to the total cumulative systems has roughly halved over time from just under 1 % in 2010 to 0.40 % in 2025.

Data: IRENA 2026. Graph: PSE Projects GmbH 2026. Date of data: 04/2026

Annually Installed PV System Capacity in Germany

Percentage of Annual Capacity by System Size



The annual distribution of the different system size categories strongly depends on current:

- regulations
- market incentives (like EEG)
- tender procedures
- bankability (trust of investors)

Note:

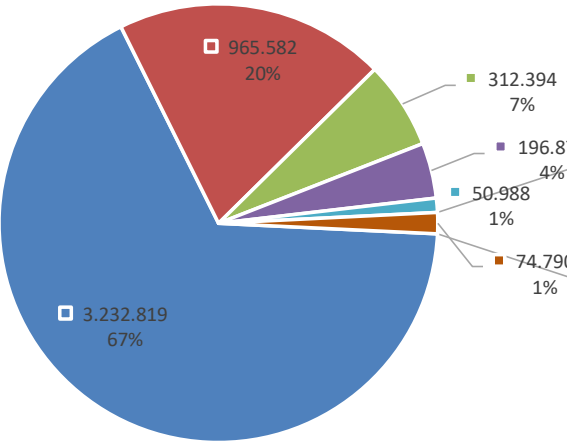
“Building” includes roofs, facades and plug-in systems.

“Ground-mounted” includes bodies of water, parking lots and other structures.

Number of Total Cumulative PV Installations by System Size in 2024

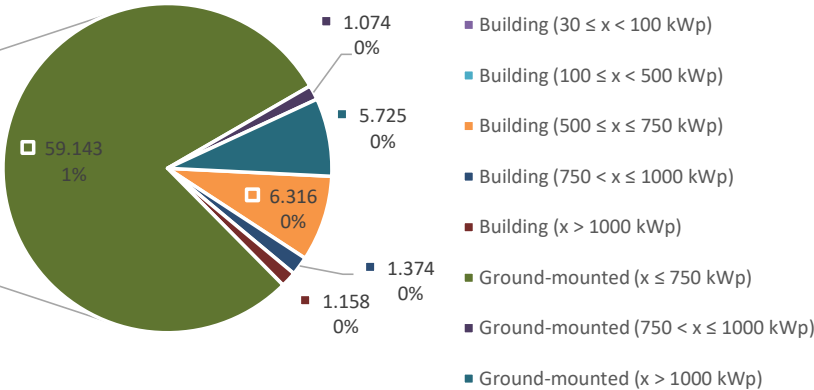
Germany

Total Installations and Share of Grid-Connected PV Systems



© Fraunhofer ISE

Total Installations and Share of PV Systems > 500 kWp



- Building (x ≤ 10 kWp)
- Building (10 < x < 20 kWp)
- Building (20 ≤ x < 30 kWp)
- Building (30 ≤ x < 100 kWp)
- Building (100 ≤ x < 500 kWp)
- Building (500 ≤ x ≤ 750 kWp)
- Building (750 < x ≤ 1000 kWp)
- Building (x > 1000 kWp)
- Ground-mounted (x ≤ 750 kWp)
- Ground-mounted (750 < x ≤ 1000 kWp)
- Ground-mounted (x > 1000 kWp)

At the end of 2024, about 4.8 million grid-connected PV systems were installed in Germany.

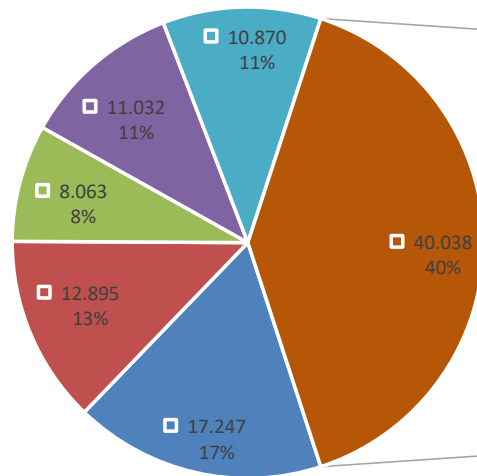
Note:
In 2024, around 800,000 plug-in systems (up to 800 W feed-in power from so-called balcony PV systems), were registered in Germany. Due to underreporting, the actual number is estimated to be around 3 million installed systems. [1]

© Fraunhofer ISE

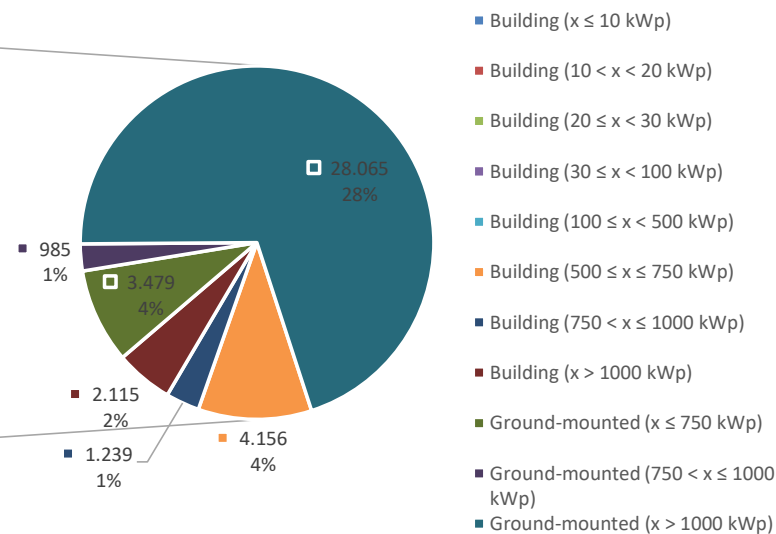
Total Cumulative Capacity of PV Installations by System Size in 2024

Germany

PV Capacity (in MW_p) and Share of All Grid-Connected PV Systems



PV Capacity (in MW_p) and Share of All Systems >500 kWp



At the end of 2024, total cumulative PV capacity in Germany amounted to around 100.1 GW_p.

© Fraunhofer ISE

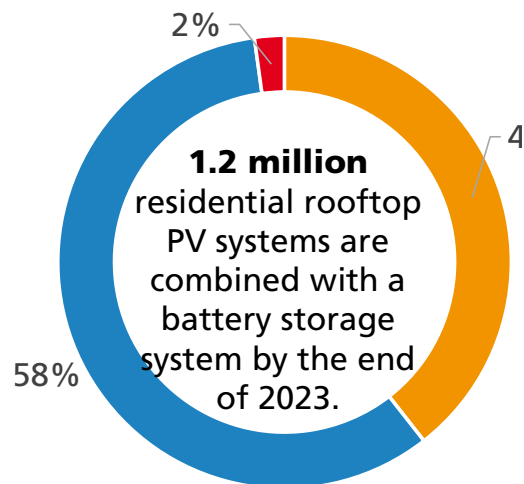
Note:

Balcony solar systems (up to 800 W feed-in power) accounted for 720 MW_p in 2024. These were mostly within the building PV system class ($x \leq 10$ kW_p), [1].

Share of Residential Rooftop Systems with and without Battery Storage

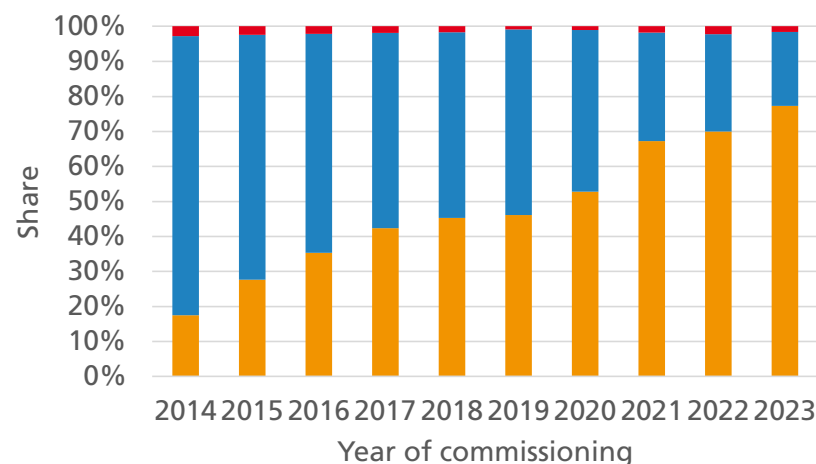
Germany

Share at the End of 2023



■ PV systems with battery storage

Share in Year of Commissioning



■ PV system without battery storage

■ Unknown

By the end of 2023, over 1.2 million units, or 40 percent of all residential PV systems have a battery energy storage system (BESS).

The share of commissions for residential rooftop PV systems with BESS increased from <20% in 2014 to nearly 80% in 2023.

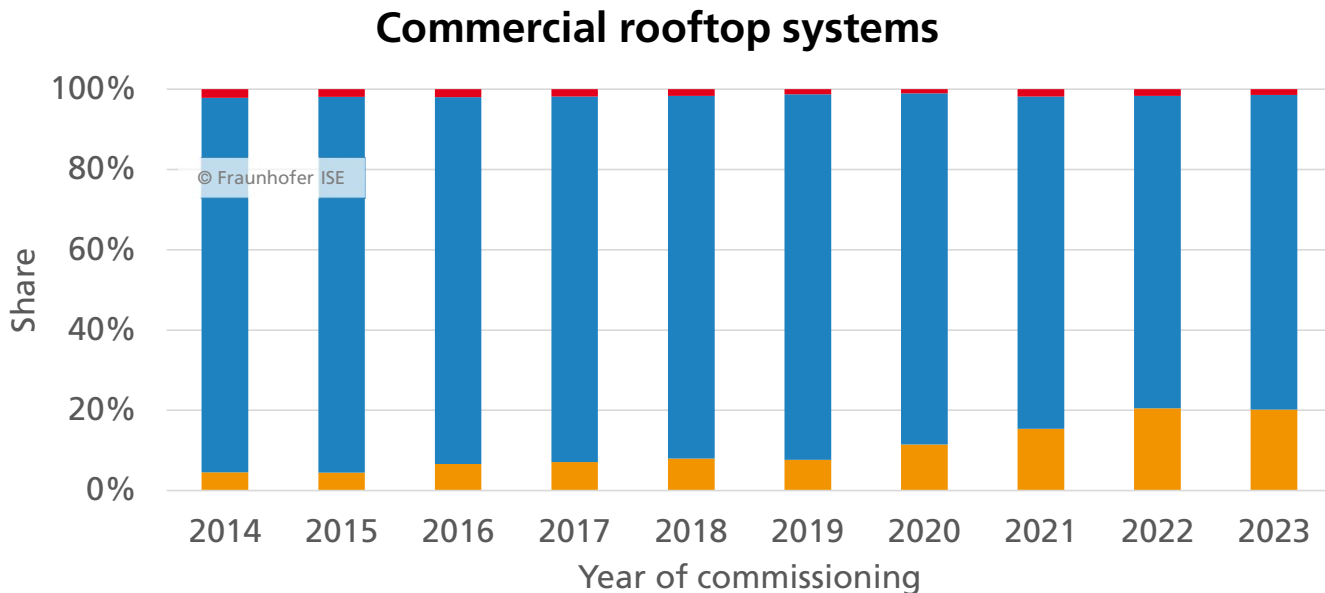
© Fraunhofer ISE

Note:

Residential rooftop PV systems are defined as all systems on buildings with a maximum capacity of 30 kWp according to MaStR-Data.

Share of PV Installations with and without Battery Storage, BESS

Commercial Rooftop and Utility-Scale Systems in Germany



The share of commissions for PV installations with BESS increased from 5% in 2014 to 20% in 2023 in the commercial rooftop sector.

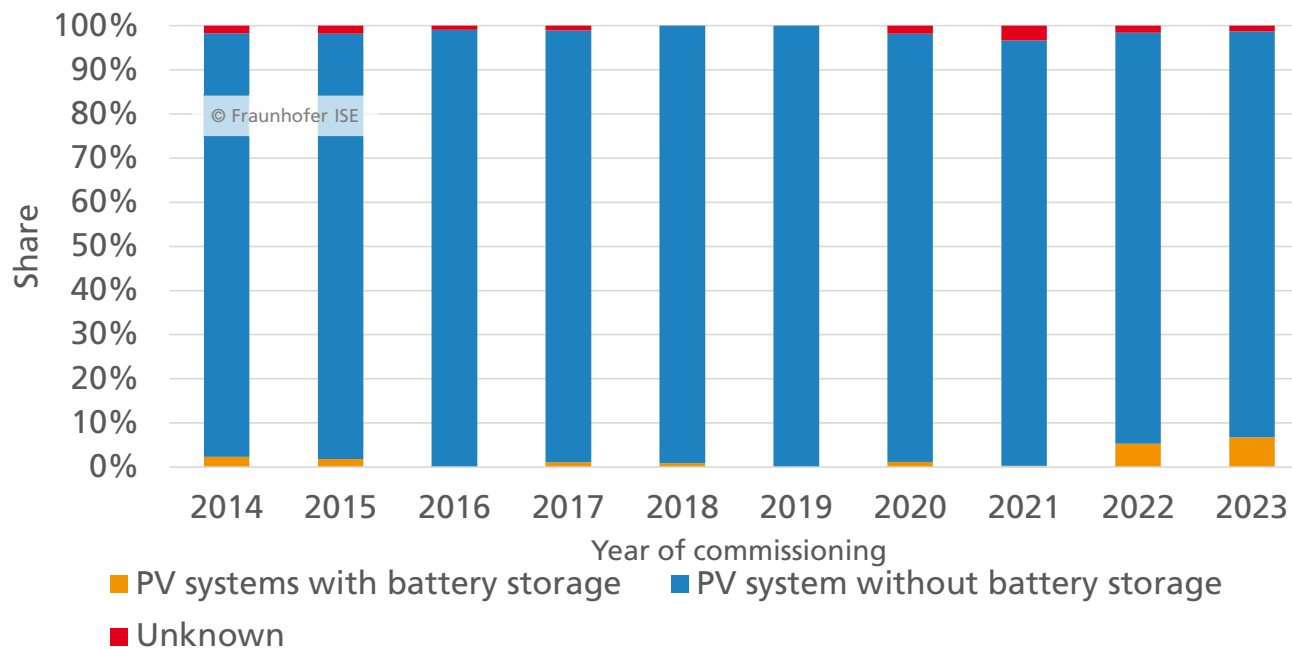
■ PV systems with battery storage ■ PV system without battery storage ■ Unknown

Note: Commercial rooftop PV systems are defined as all systems on buildings with a capacity greater than 30 kWp according to MaStR-Data.

Share of PV Installations with and without Battery Storage, BESS

Commercial Rooftop and Utility-Scale Systems in Germany

Utility-scale ground-mounted systems

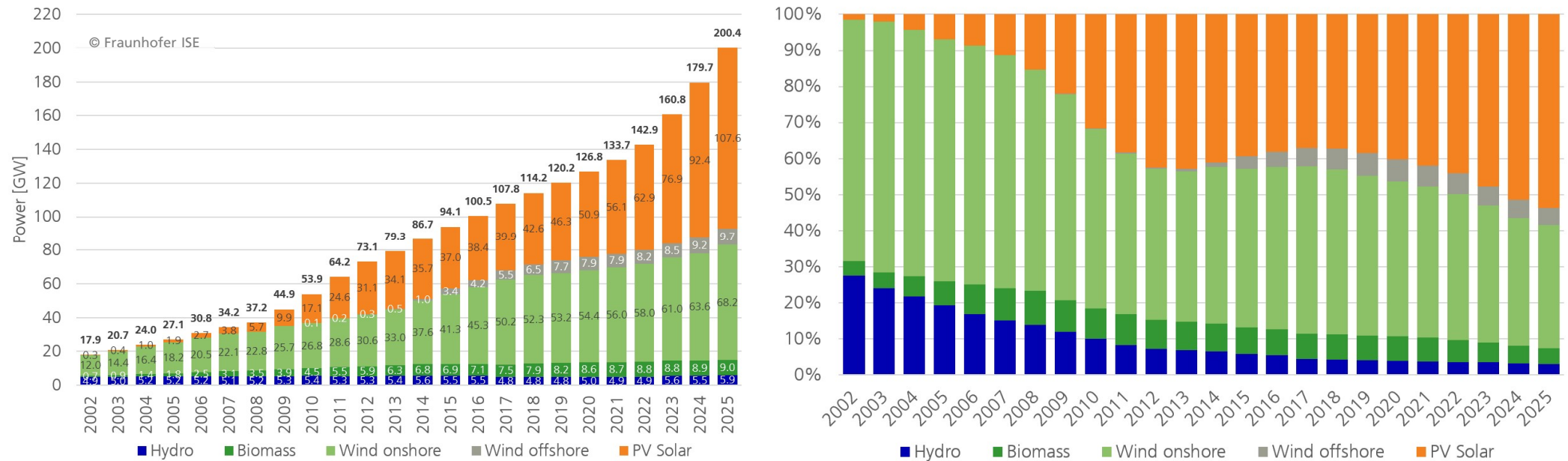


In 2022 and 2023 there was a significant increase in commissions for PV with BESS, reaching a share of around 7% in the utility-scale PV power plant sector.

Note: Utility-scale ground-mounted PV systems are defined as all ground-mounted systems with a capacity greater than 1 MWp according to MaStR-Data.

Electrical Capacity of Renewable Energy Sources (RES)

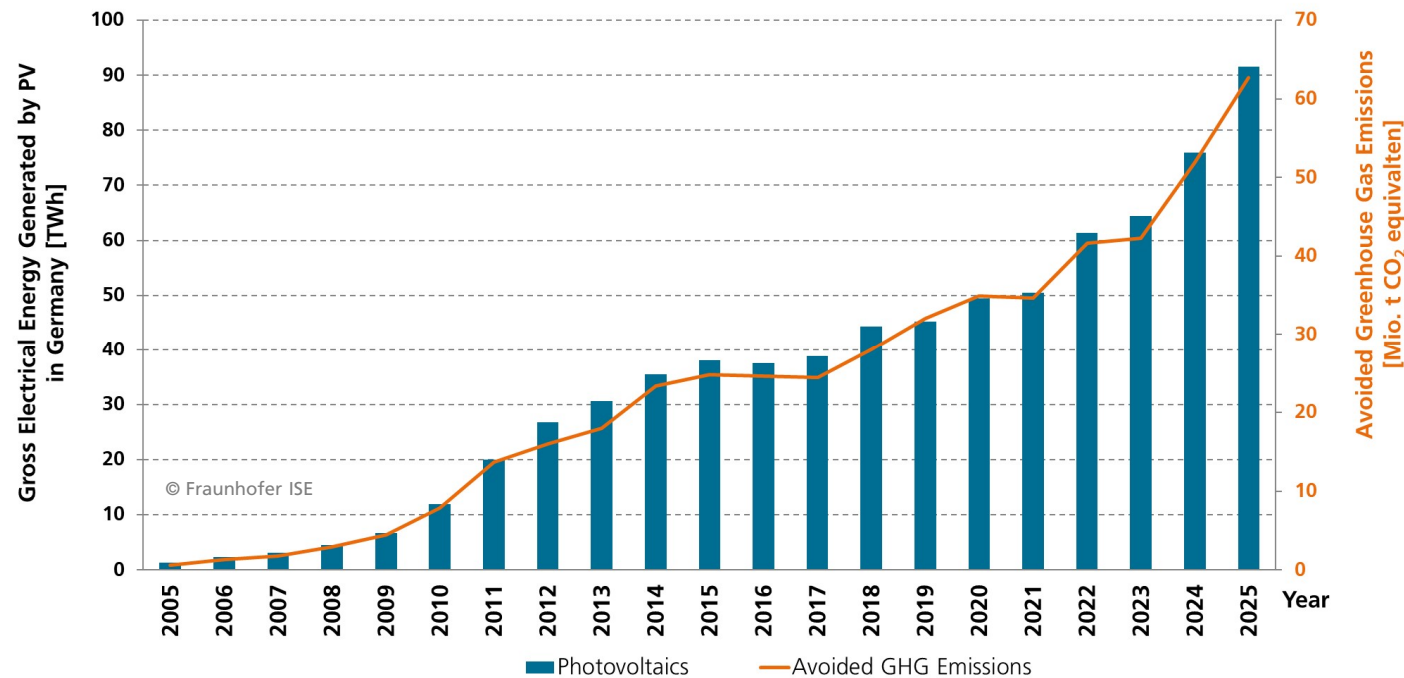
Germany



Renewable energy sources accounted for 200.4 GW of the total 272.0 GW net installed electricity generation capacity in Germany in 2025, resulting in a RE share of 73.7% of total capacity.

PV Energy Generated and Resulting GHG Emissions Avoided

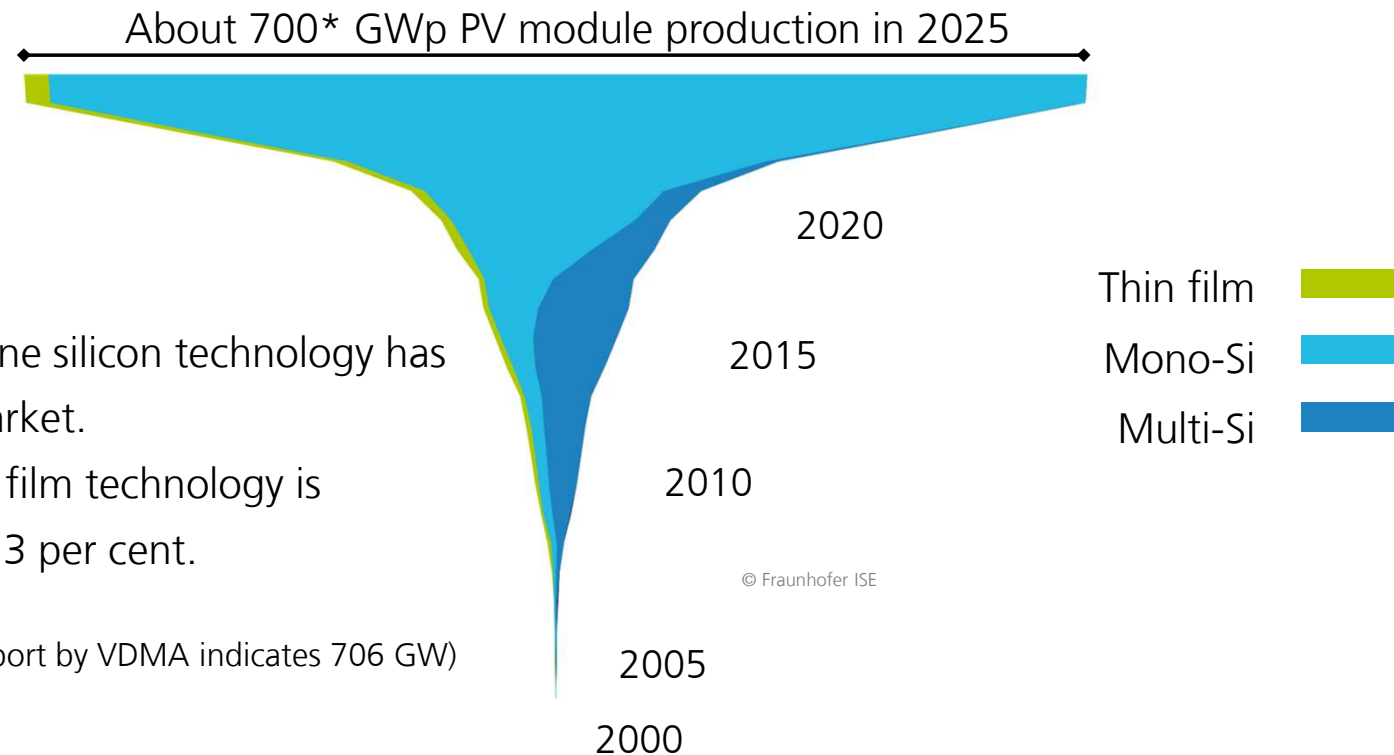
Germany



In 2025, greenhouse gas emissions of about 62.7 Mio. t CO₂-equivalent were avoided due to 91.6 TWh PV electricity consumed in Germany.

Annual PV Production by Technology

Worldwide (in GWp)

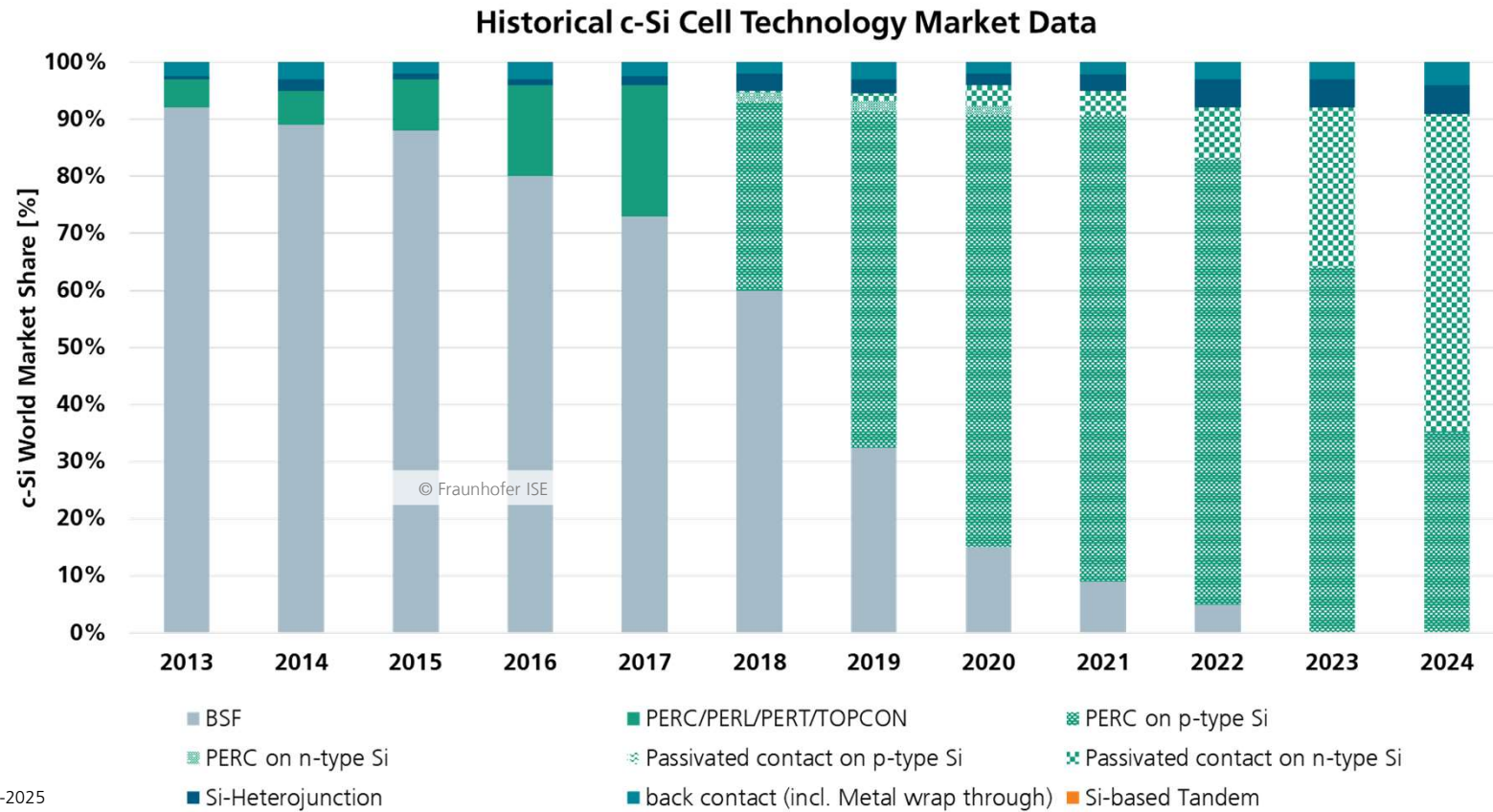


- In practice, multi-crystalline silicon technology has disappeared from the market.
- The market share of thin film technology is currently between 2 and 3 per cent.

*Estimated number (ITRPV 17th report by VDMA indicates 706 GW)

Technology Overview

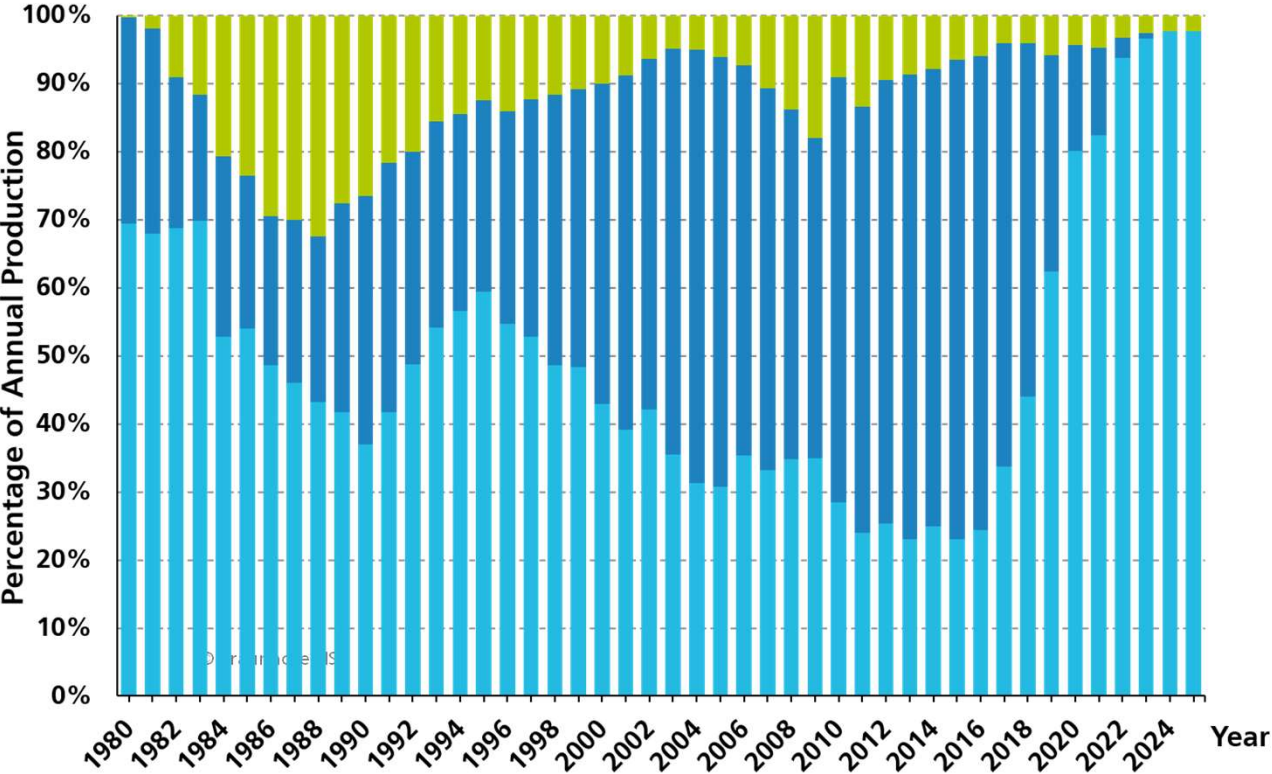
Different crystalline-Silicon Cell Technology Market Shares



Source: based on ITRPV 2013-2025

PV Production by Technology

Percentage of Global Annual Production



Production 2025 (GWp)

Thin film*	16
Multi-Si	0
Mono-Si	690
Total	706 (ITRPV)

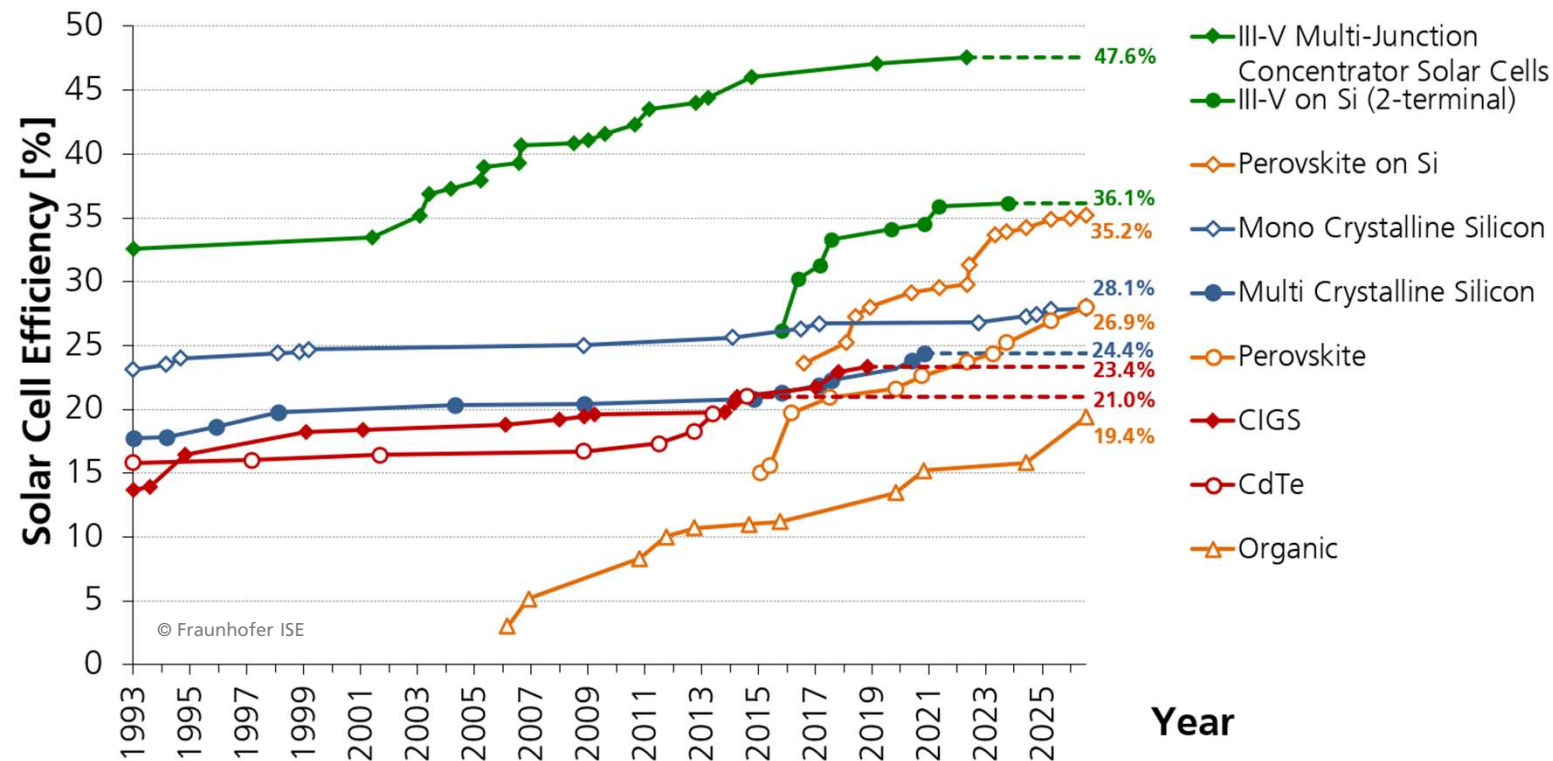
*only First Solar is considered when calculating the thin film technology.

Data: from 2000 to 2009: Navigant; from 2010 to 2021 IHS Markit; from 2022 estimates based on IEA and other sources. Graph: PSE Projects GmbH 2026 . Date of data: 07/2026

2. Solar Cells / Modules / System Efficiency

- Development in the Laboratories
- Development in the PV Industry
- Performance Ratio (PR)

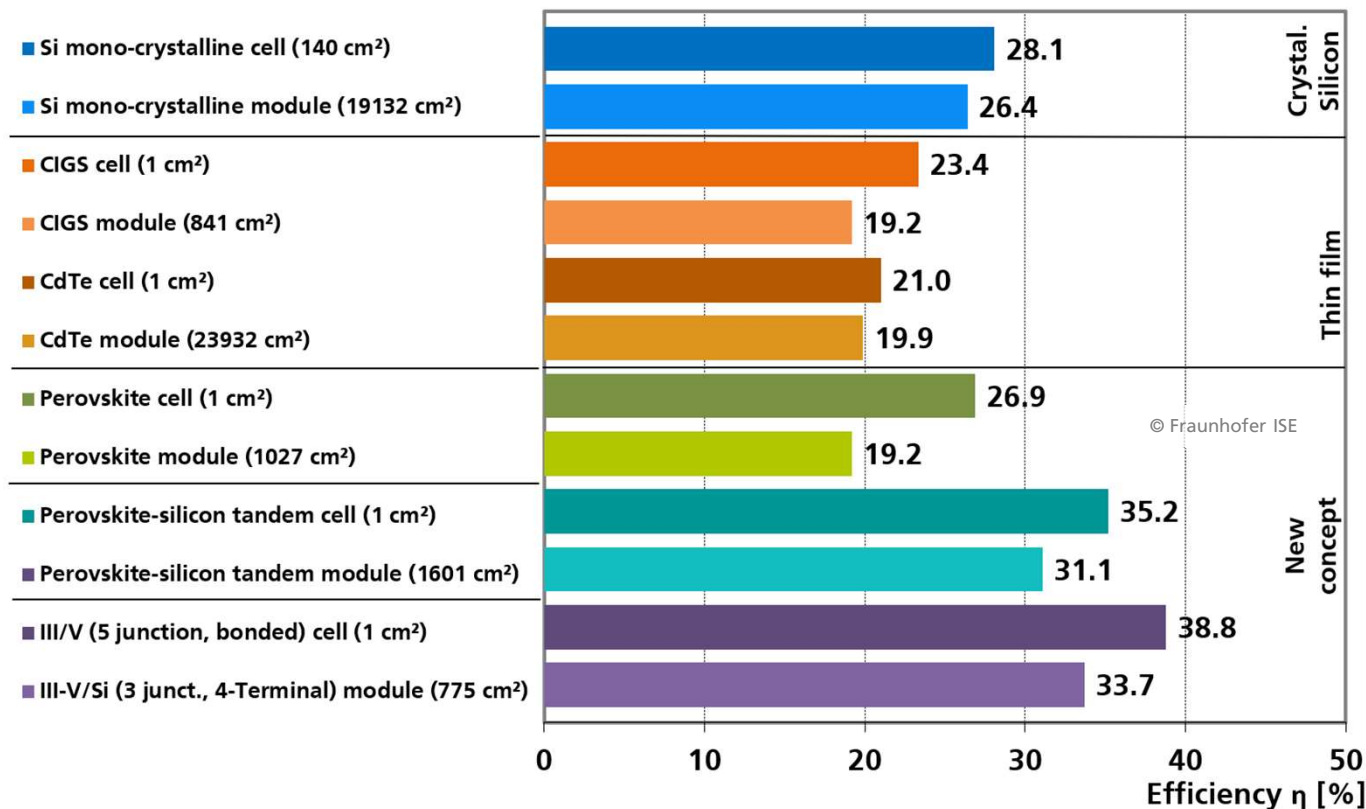
Development of Laboratory Solar Cell Efficiencies



Data: Solar Cell Efficiency Tables (Versions 1 to 68), Progress in Photovoltaics: Research and Applications, 1993-2026. Graph: Fraunhofer ISE 2026. Date of data: 06/2026

Efficiency Comparison of Technologies

Best Lab Cells vs. Best Lab Modules



Note: In mass production, the cell-to-module ratio (CTM) improved in past years by reducing losses and using possible gains when integrating solar cells in modules.

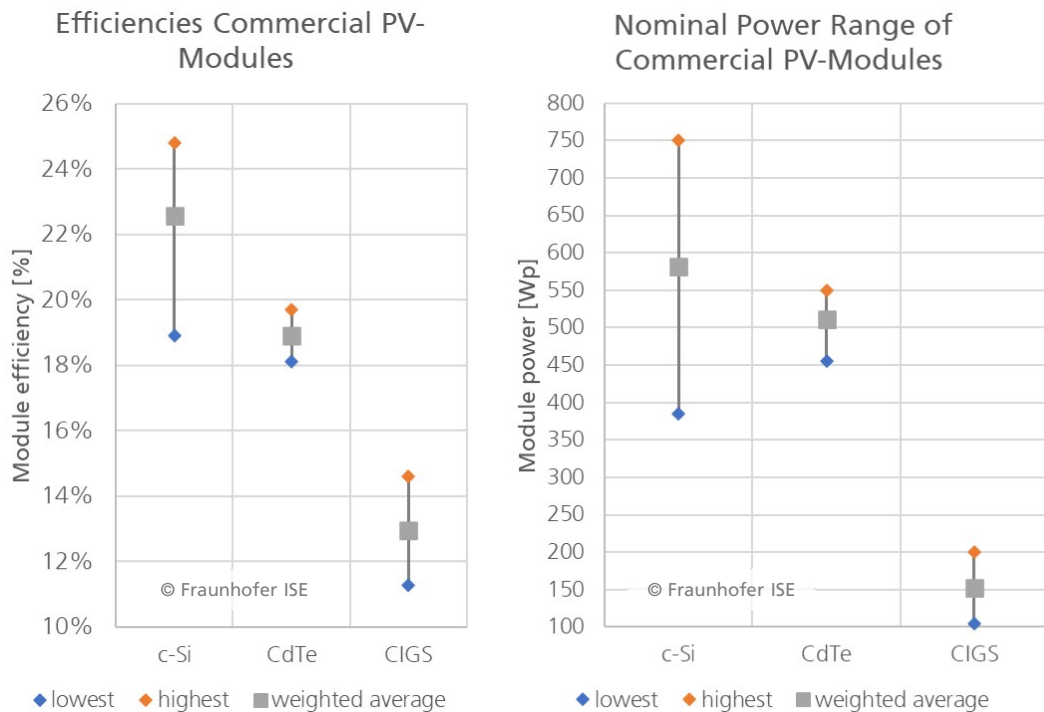
Fraunhofer ISE provides the software suite SmartCalc.CTM for determining precise Cell-to-Module (CTM) power loss analyses. Geometrical losses, optical losses and gains as well as electrical losses are considered in the analysis.

www.cell-to-module.com

Data: Green et al.: Solar Cell Efficiency Tables (Version 68), Progress in PV: Research and Applications 2026. Graph: PSE Projects GmbH 2026. Date of data: 06/2026

Current Efficiencies and Power of Commercial PV Modules

Sorted by Technology



- Total weighted average efficiency of crystalline Silicon (c-Si) wafer-based modules is 22.7% in Q4-2024 (21.6% in Q4-2023); weighting factor is total shipments in year 2024. Lowest module efficiency in this group is 18.9% (17.4% one year before) and highest value is 24.8% (23.3% in 2023).
- Top 10 manufacturers represent about 85% of total shipment volume in 2024 and origin mainly in Asia.
- n-type TopCon and Heterojunction replaces p-type PERC technology.

Note: The selection is based on modules from the top 10 manufacturers in 2024, with module data sheets available worldwide at the end of September 2025. For CdTe, only data from First Solar and for CIGS technology, only modules from Avancis have been considered due to limited number of suppliers. Some products for Building-Integrated PV (BIPV) have not been considered.

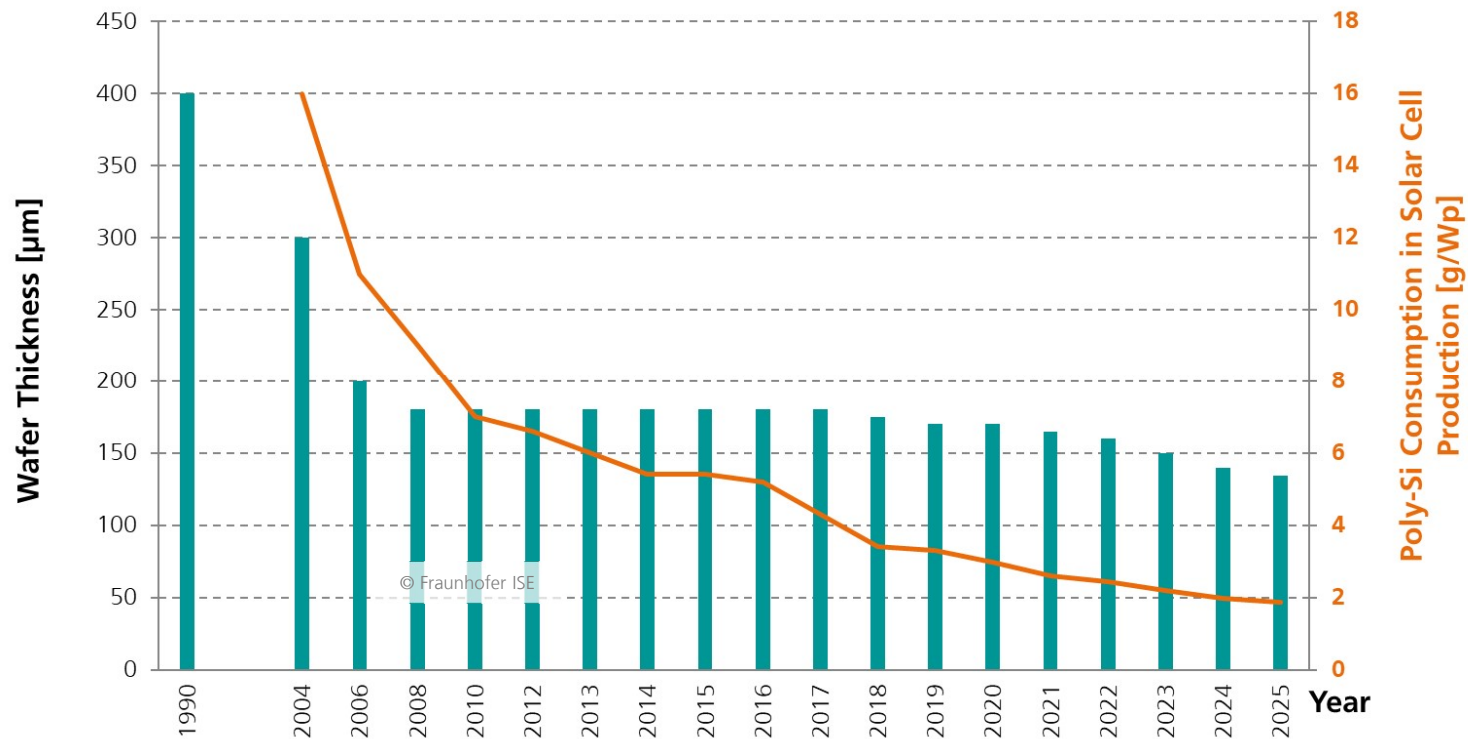
Data Source: company product data sheets; Graph: PSE Projects GmbH 2025; Date of data: 09/2025

3. Life Cycle Assessment (LCA) and Sustainability Aspects

- Silicon usage, wafer thickness and kerf loss for c-Si
- Energy Payback Time EPBT: Development and comparison

c-Si Solar Cell Development

Wafer Thickness [μm] & Silicon Usage [g/Wp]



Polysilicon consumption in solar cell production (in grams per watt-peak) has decreased by almost 10% year-on-year since 2004 due to:

- reductions in wafer thickness
- kerf loss, and
- by process optimizations such as recycling silicon from kerf loss
- Progress in cell efficiency also has impacted the specific silicon usage.

Data: until 2012: EU PV Technology Platform Strategic Research Agenda, from 2012: ITRPV; from 2016 ISE without recycling of Si; from 2017 ongoing with recycling of Si. Graph: PSE Projects GmbH 2025

Energy Payback Time: Historical Trend

Harmonized Study Data for Monocrystalline Silicon Rooftop PV Systems

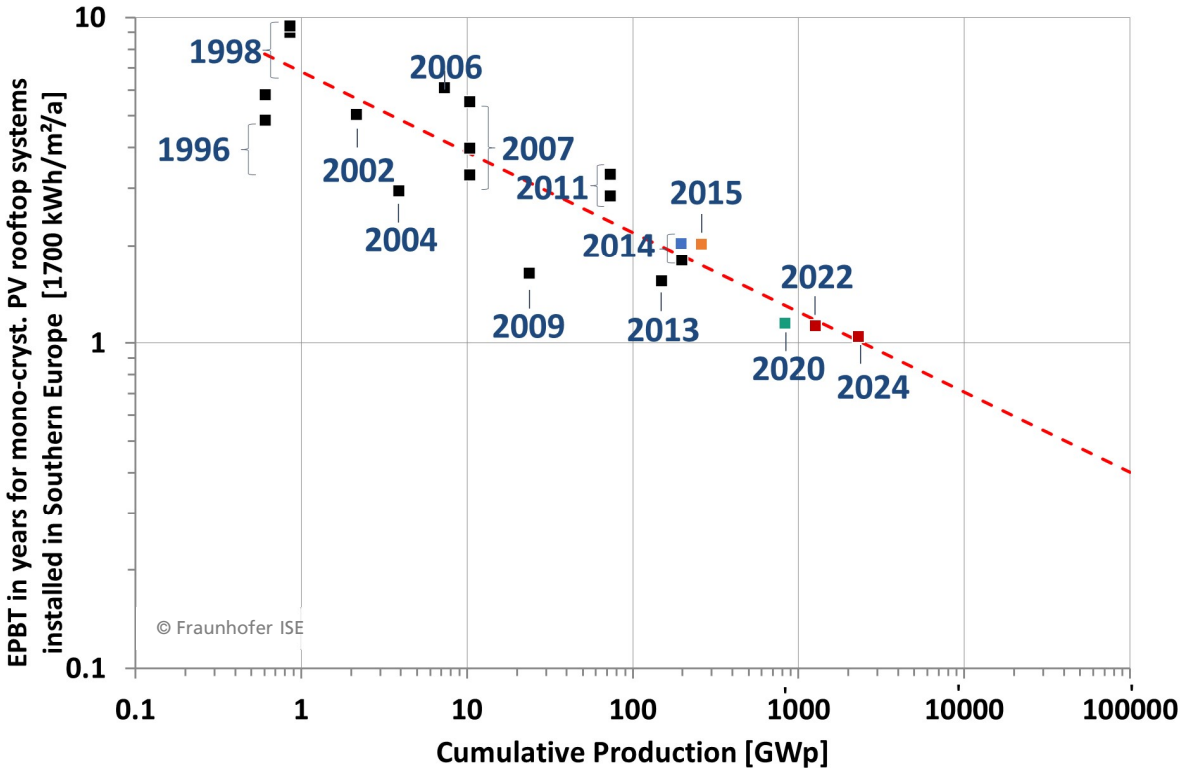
Learning Rate:
Each time the cumulative production doubled, the EPBT went down by 15.7% over the last 24 years.

Harmonization methodology
based on Koppelaar (2016) harmonized results and harmonization parameters

1) Performance Ratio (PR)
based on average annual PV yield during lifetime
PV system lifetime (years) 25
Degradation rate (average p.a.) 0.70%
PR (initial) 80%
PR (incl. average degradation during lifetime) 73.6%

2) Grid efficiency
for converting PV yield in primary energy equivalents
Grid efficiency 35%

EPBT of Leccisi (2016), Louwen (2014), Friedrich (2020), and Subasi (2022 & 2024) were harmonized with 1) PR (incl. average degradation) and 2) grid efficiency to results of Koppelaar (2016)*



Data and Graph: Fraunhofer ISE 2025; adapted from Subasi et. al. (2025) - PV Module Energy Payback Time Analysis for Silicon Solar Cell Technologies, EU PVSEC 2025, Bilbao, Spain

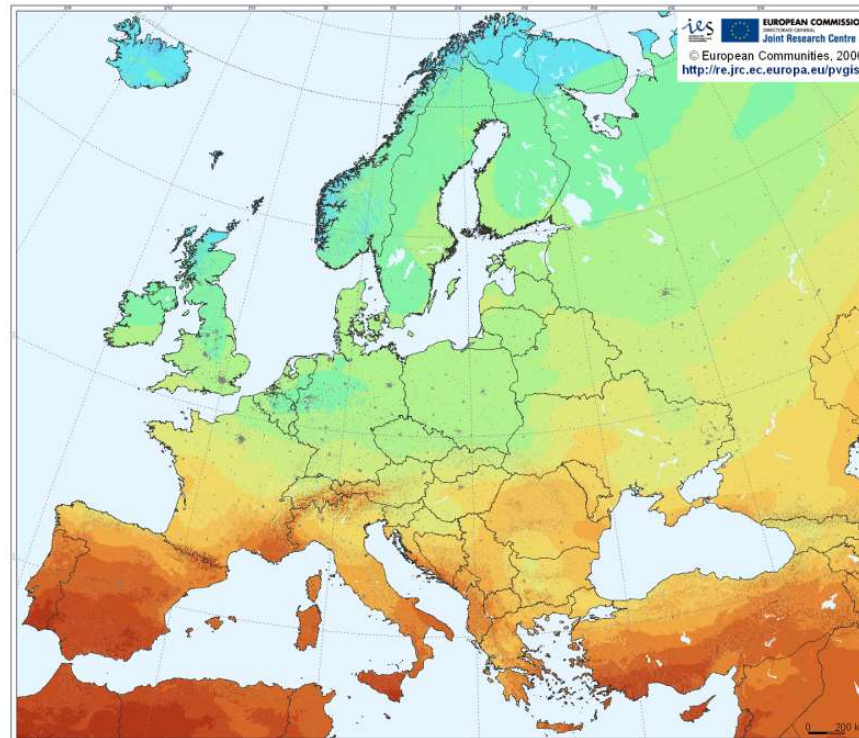
Irradiation: 1700 kWh/m²/a at an optimized tilt angle; **Years:** Estimated average year of original data.
- - - Learning Rate taking into account efficiency improvements.

*■ Koppelaar (2016) - Solar-PV energy payback and net energy: Meta-assessment of study quality, reproducibility, and results harmonization, Renewable and Sustainable Energy Reviews
■ Leccidi et al. (2016) - The Energy and Environmental Performance of Ground-Mounted Photovoltaic Systems—A Timely Update, Energies
■ Louwen et al. (2014) - Life-cycle greenhouse gas emissions and energy payback time of current and prospective silicon heterojunction solar cell designs, Progress in Photovoltaics
■ Friedrich et al. (2020) - Global Warming Potential and Energy Payback Time Analysis of Photovoltaic Electricity by Passivated Emitter and Rear Cell (PERC) Solar Modules, submitted JPV
■ Subasi et. al. (2025) - PV Module Energy Payback Time Analysis for Silicon Solar Cell Technologies, EU PVSEC 2025, Bilbao, Spain

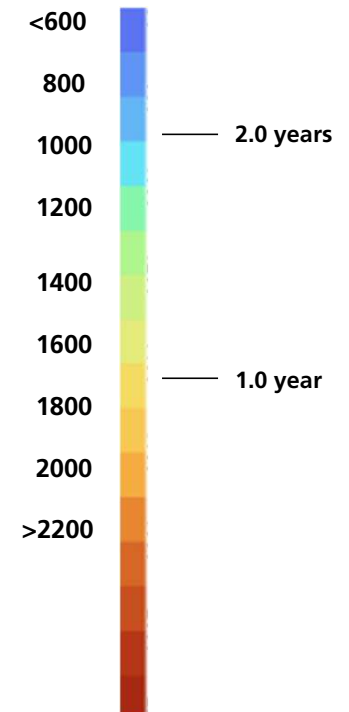
Energy Pay-Back Time of Silicon PV Rooftop Systems

Geographical Comparison

- Rooftop PV system using mono-crystalline silicon cells* produced in China
- EPBT is dependent on irradiation, but also on other factors like grid efficiency**.
- In addition to solar radiation, grid efficiency also has an influence on EPBT. A higher grid efficiency increases the EPBT.



Irradiation (GTI, kWh/m²/a) EPBT***



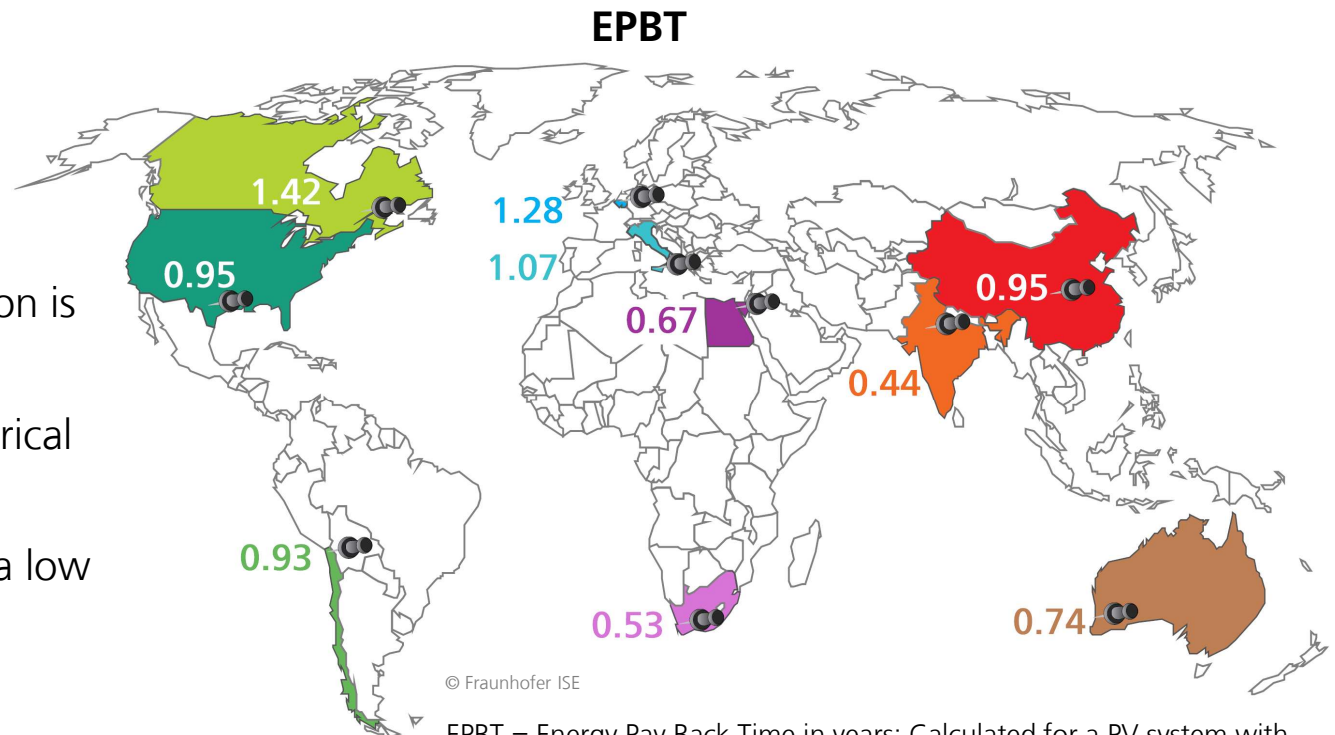
Data source: Fraunhofer ISE. Image: JRC European Commission. Graph: PSE Projects GmbH 2020 (Modified scale with updated data from Fraunhofer ISE)

World Map EPBT of Silicon PV Rooftop Systems

Comparison of EPBT China

Influencing Factors and Interpretation

- **EPBT:** the lower, the better
- **Irradiation:** the higher, the better
- **Grid efficiency:** the higher, the better
 - in countries where upstream production is located (better energy mix to generate electrical power, less losses in the electrical transmission network), and
 - at downstream (where PV is installed) a low grid efficiency reduces the EPBT.

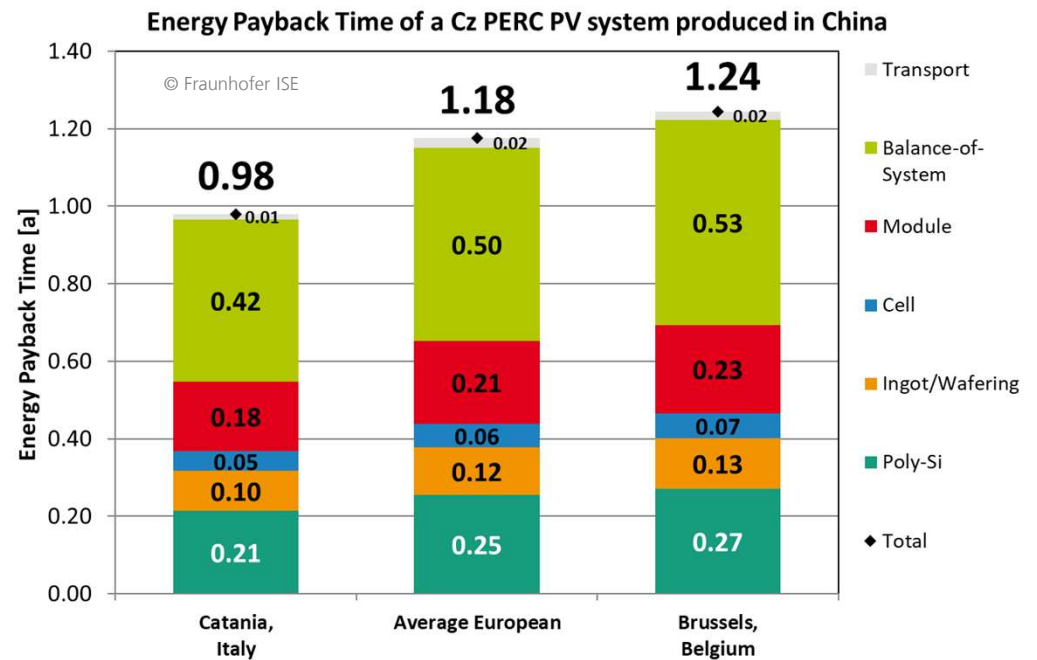
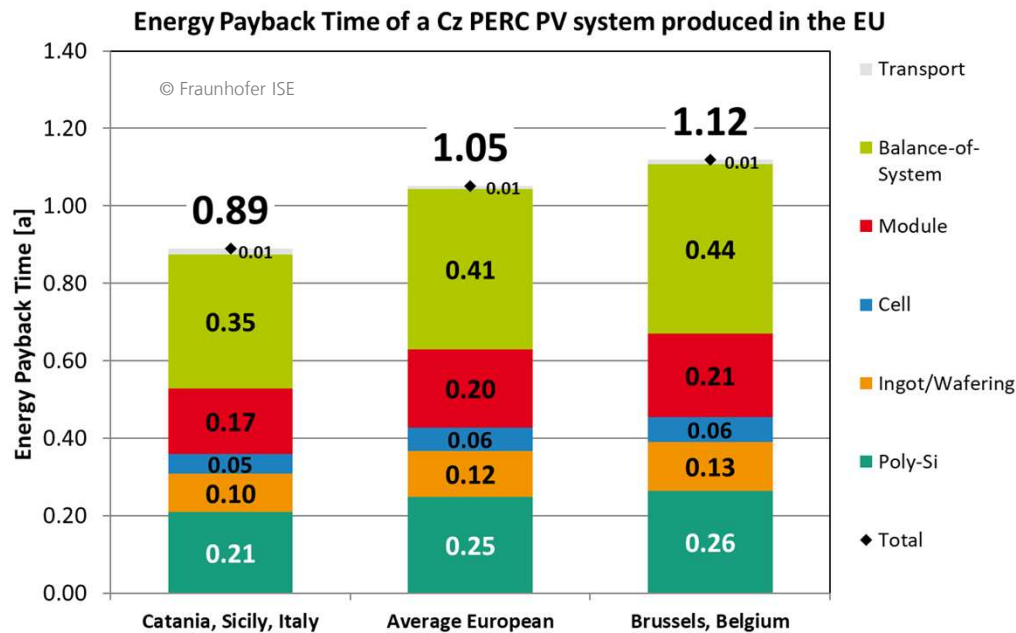


EPBT = Energy Pay Back Time in years: Calculated for a PV system with Cz PERC 60-cell modules with 19.9 % efficiency, produced in China

Data source: Fraunhofer ISE. Calculations for year 2021.

Energy Payback Time of Silicon PV Rooftop Systems

Comparison of EPBT China / EU, Local Irradiation and Grid Efficiency in 2021

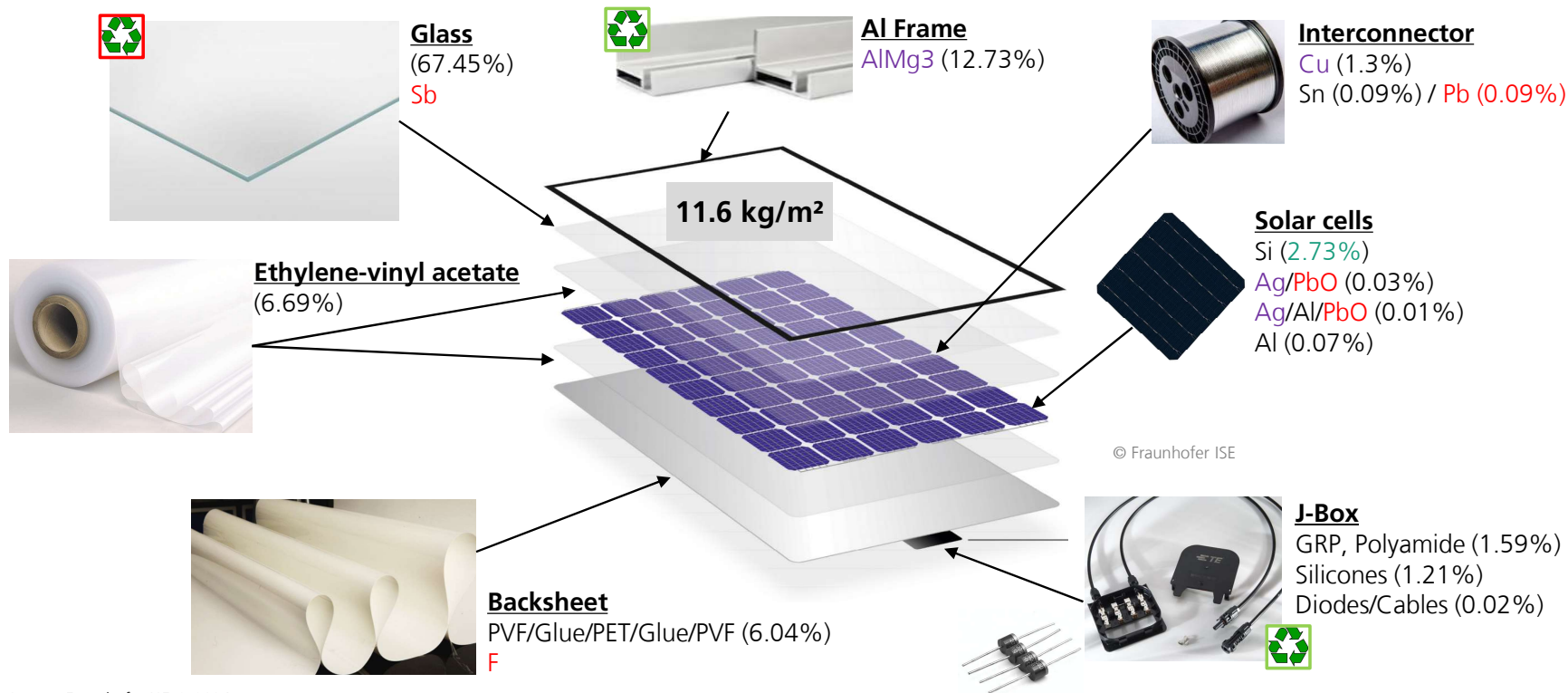


EPBT for PV systems produced in Europe is shorter than for those produced in China because of better grid efficiency in Europe.

Data source: Fraunhofer ISE. Calculations for year 2021 made at 22-July 2022

PV Module

Materials and Components



Please note: Highly transparent glass can also be produced without antimony (Sb), and some European suppliers are doing so. It is technically feasible to recycle and reuse almost 100% of the materials used in PV modules. The European WEEE Directive stipulates that at least 80% of the module mass of old modules must be processed and recycled for reuse. For economic reasons, however, only the glass, frame and junction box (J-Box) are recycled today.

Color legend:

Available/harmless materials

Rare/valuable materials

Hazardous substances



Recycling takes place



Downcycling takes place

Source: Fraunhofer ISE © 2024

PV Module

Measures to Improve the Life Cycle

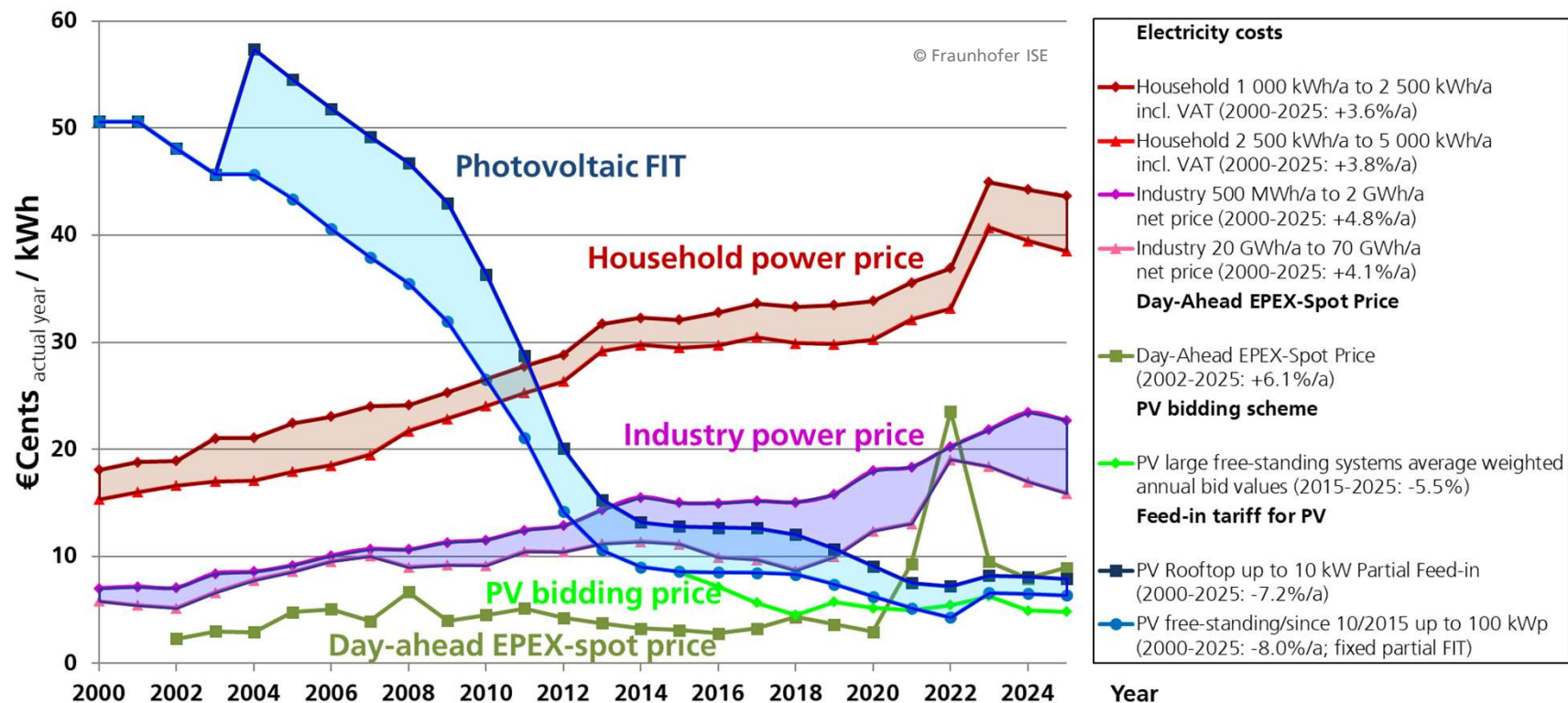
What to reduce or avoid?	What solutions are available?
Material consumption (general)	Higher efficiency PV modules; longer lifetime; better recyclability; bio-based polymers
Silicon	Thinner wafer
Silver (Ag)	Multi-busbars with more precise application of the silver paste; busbar-less cells; Replacing silver (Ag) by copper (Cu) or aluminium (Al)
Antimony (Sb) in glass	Sb-free solar glass
Fluorinated backsheets	Double-glass modules; alternative backsheets without fluorine compounds
Glass consumption	Thinner glass, reuse or recycling of solar glass instead of downcycling
Lead (heavy metal issues)	Lead-free solders or alternative contacting; alternative interconnection using electrically conductive adhesive (ECA)
Aluminium frame	Frameless modules; steel or plastic frame
Module failure due to faulty bypass diode	Install replaceable bypass diode

4. Price Development

- Electricity costs
- Market incentives in Germany
- Costs for PV systems
- Price Learning Curve

Electricity Prices, PV Feed-In Tariffs (FIT) and Bidding Scheme in Germany

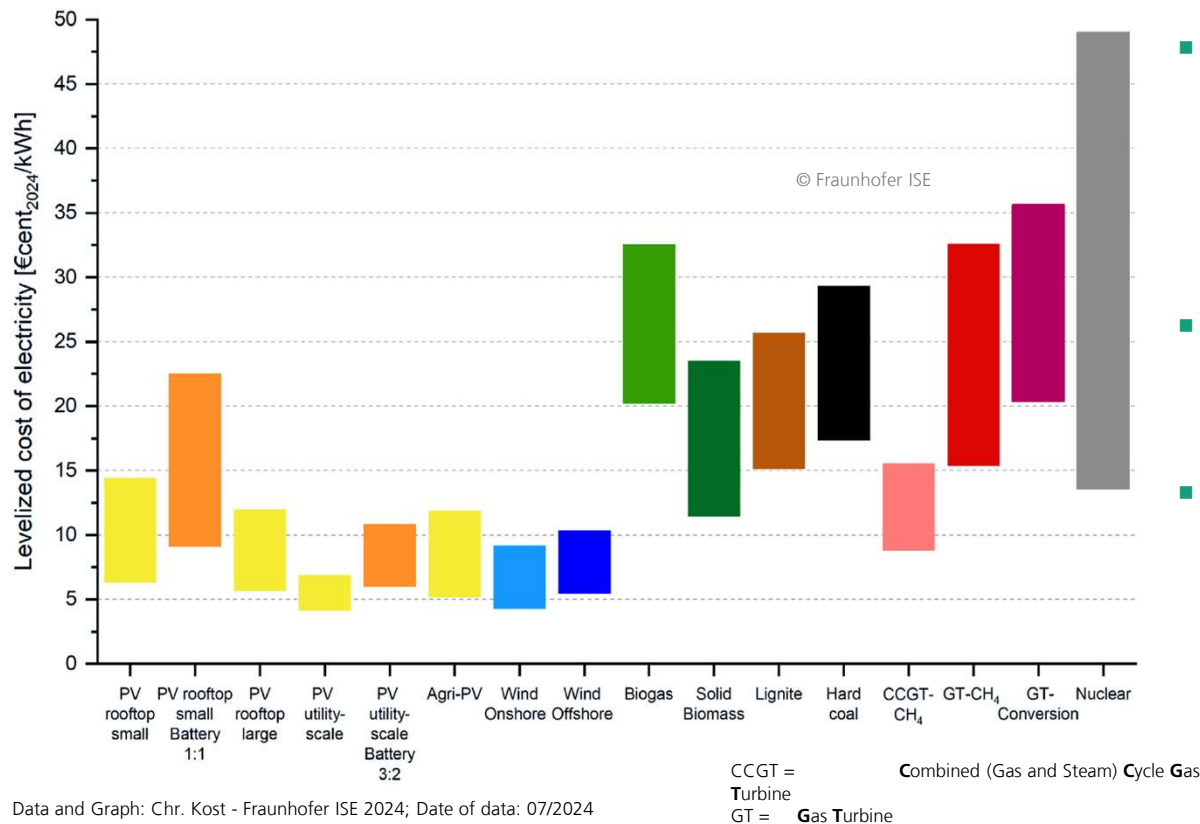
With Photovoltaic Rooftop Systems Partial Feed-In Tariff



Data: BNA; Destatis; energy-charts.info; Design: B. Burger - Fraunhofer ISE. Graph: PSE Projects GmbH 2026; Date of data: 05/2026

Electricity Generation Costs (LCOE)

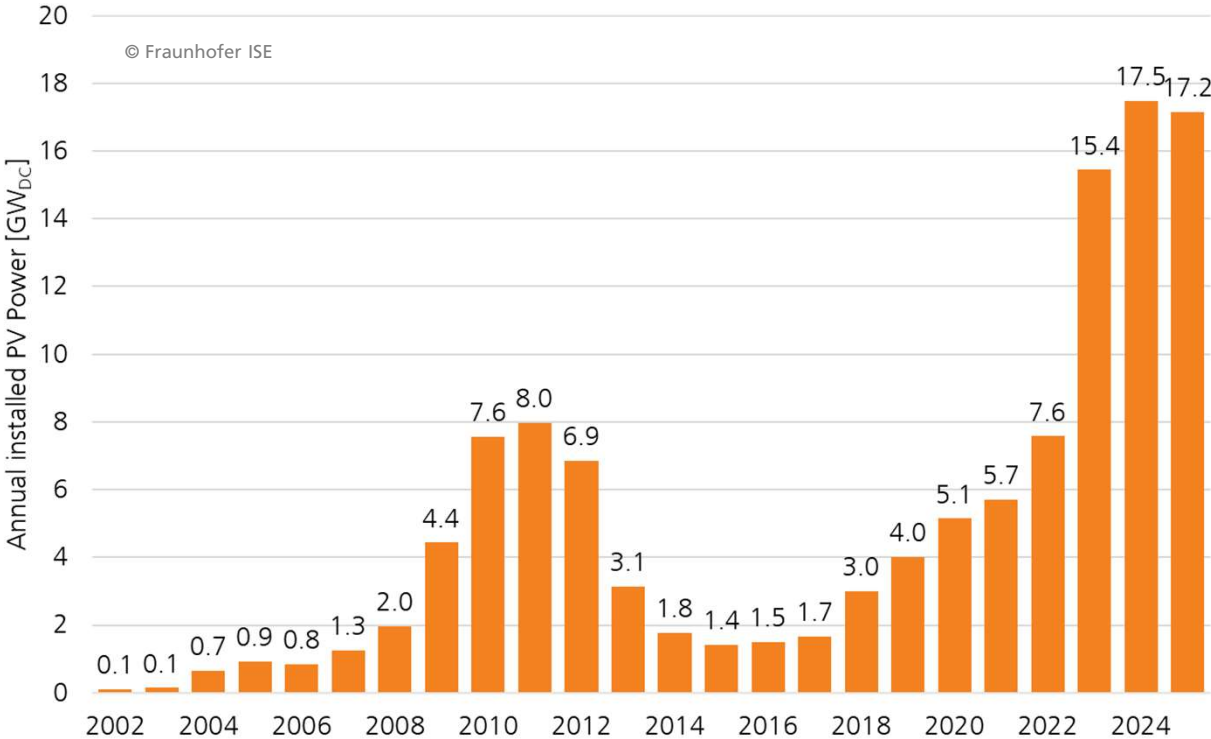
Comparison of electricity generation costs in €cent/kWh for various energy sources in Germany



Data and Graph: Chr. Kost - Fraunhofer ISE 2024; Date of data: 07/2024

- With costs ranging from 4.1 to 9.2 cents per kilowatt hour, ground-mounted PV systems and onshore wind turbines are the most cost-effective technologies in Germany, among both renewable and conventional power plants.
- Even solar power systems with batteries are more cost-effective than conventional power plants that run on fossil fuels.
- Hybrid solutions that combine solar and wind power with battery storage can provide cost-effective electricity from renewable sources around the clock, reducing dependence on imported fossil fuels.

PV Market Development and Incentive Schemes in Germany



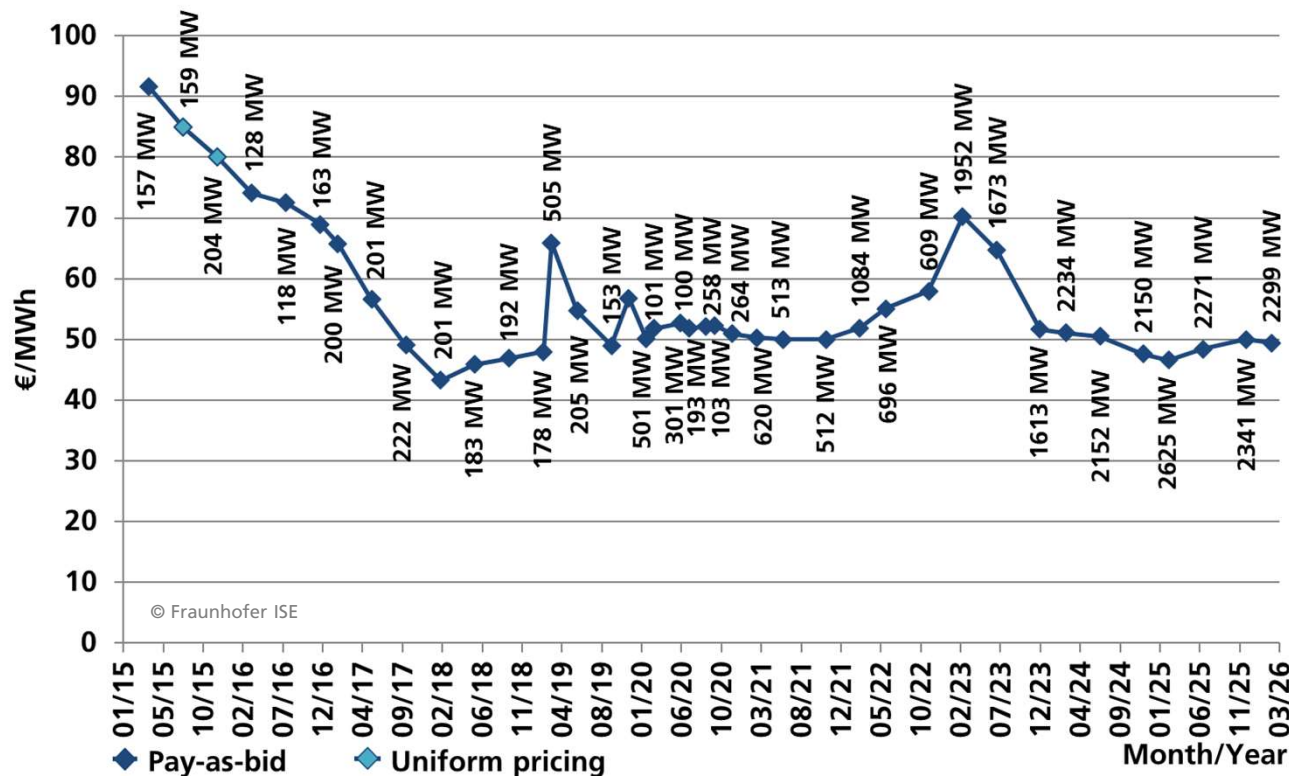
Data: BNA. Graph: B. Burger, Fraunhofer ISE Energy-Charts. Date of Data: 20.05.2026

Market Incentive	Start	End
1'000 Roofs Program	1990	1995
Cost-covering remuneration	1993	1999
100'000 Roofs-Program	1999	2003
EEG	2000	ongoing
PV Tendering scheme	2015	ongoing

The EEG 2023 law relies on a massive expansion of renewable energies with total installed PV capacity targets of 215 GW in 2030 and 400 GW in 2040. In 2025, new PV systems totaling 17.2 GW_{DC} capacity have been connected to the grid. From 2026 on, the expansion target is 20 GW of new solar installations on an annual basis.

PV Tender Scheme in Germany for Free-Standing Systems

Average, quantity weighted award value

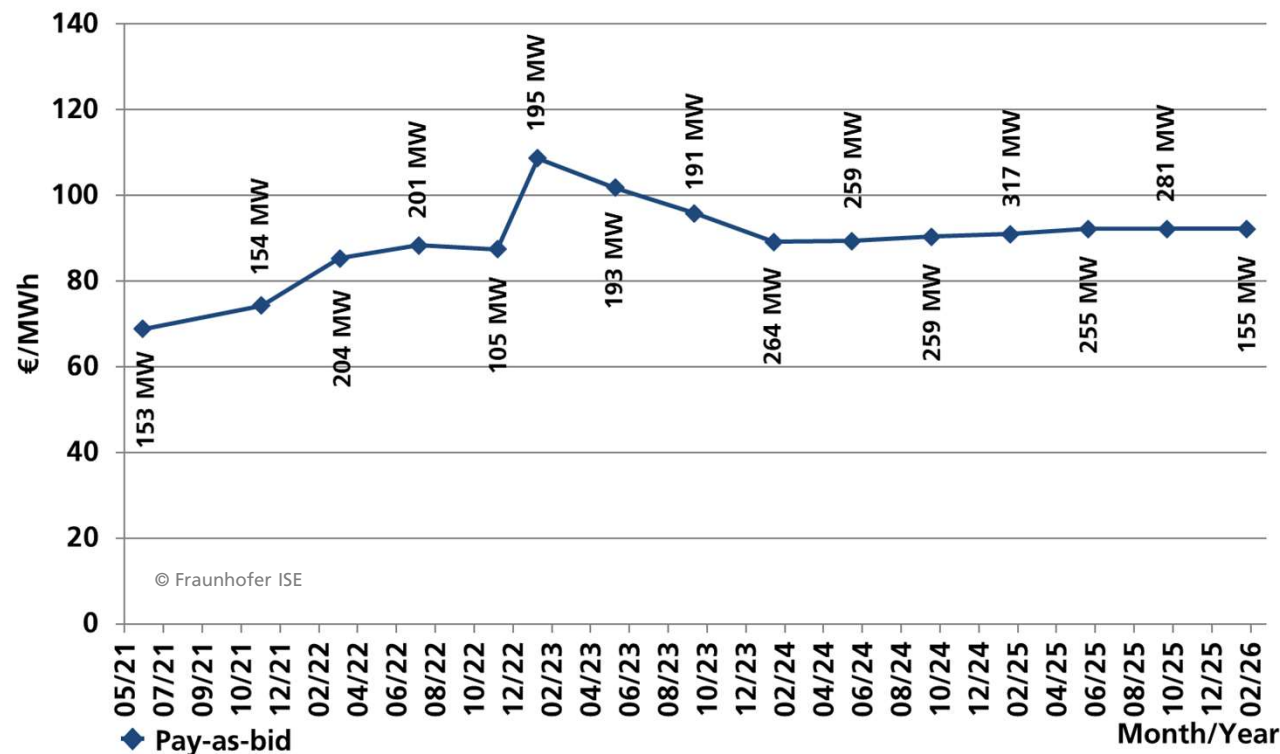


Data: BNA. Graph: PSE Projects GmbH 2026 – Date of data: 05/2026

- The PV tender scheme for large ground-mounted systems started in April 2015. The total capacity of this scheme amounted to 28.0 GW in December 2025 with 4.83 ct€ / kWh as latest average quantity weighted award price.
- Lowest PV tender round was in February 2018 with 4.33 ct€ / kWh as average quantity weighted award price.
- PV-rooftop and special tenders are not displayed in the graph.

PV-Tender in Germany for Large Rooftop-Systems

Average, quantity weighted award value

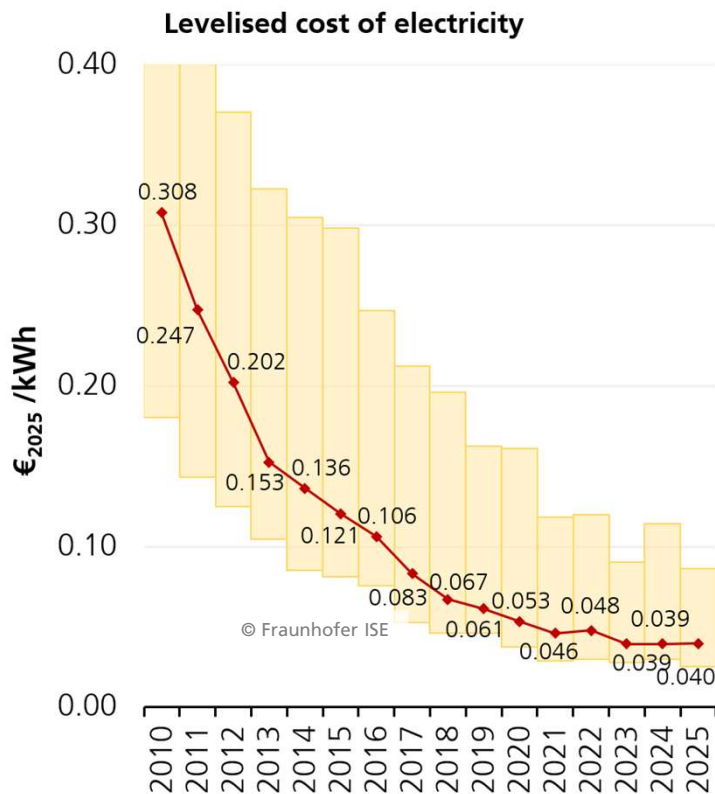


Data: BNA. Graph: PSE Projects GmbH 2026 – Date of data: 05/2026

- PV tender scheme for large rooftop systems (>750 kW) started in June 2021 and total capacity of this scheme accumulates to 2.2 GW by October 2024. The average quantity weighted award price for year 2024 was 8.97 ct€/ kWh.
- Lowest PV tender round was in June 2021 with 6.88 ct€/ kWh as average quantity weighted award price.

Global Weighted Average Levelized Costs of Electricity for Large PV Systems

(with 5th percentile and 95th percentile)

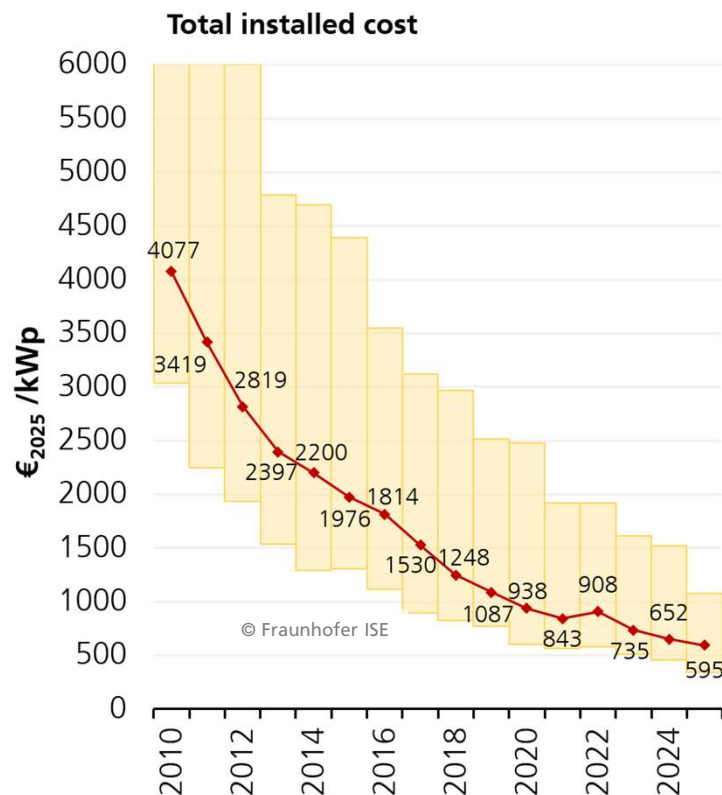


- The global weighted average LCoE for 2025 for large PV systems is 0.040 €/kWh (= 40 €/MWh).
- The 5th percentile is a value associated with the location within the data where 5% of the data is below that value. For 2025, the 5th percentile is 0.025 €/kWh (= 25 €/MWh).
- The 95th percentile is the value where 5% of the data has a higher value. For 2025 the 95th percentile is 0.086 €/kWh (= 86 €/MWh).
- Over the past 15 years, electricity generation costs of utility-scale PV systems have fallen by an average of around 12.8% compared to the previous year.

Data: IRENA (2026), Renewable Power Generation Costs in 2025, International Renewable Energy Agency, Abu Dhabi. Currency converted from USD to EUR. Date of data: July 2026
Currency converted from USD to EUR

Global Weighted Average Total Installed Costs For Large PV Systems

(with 5th percentile and 95th percentile)

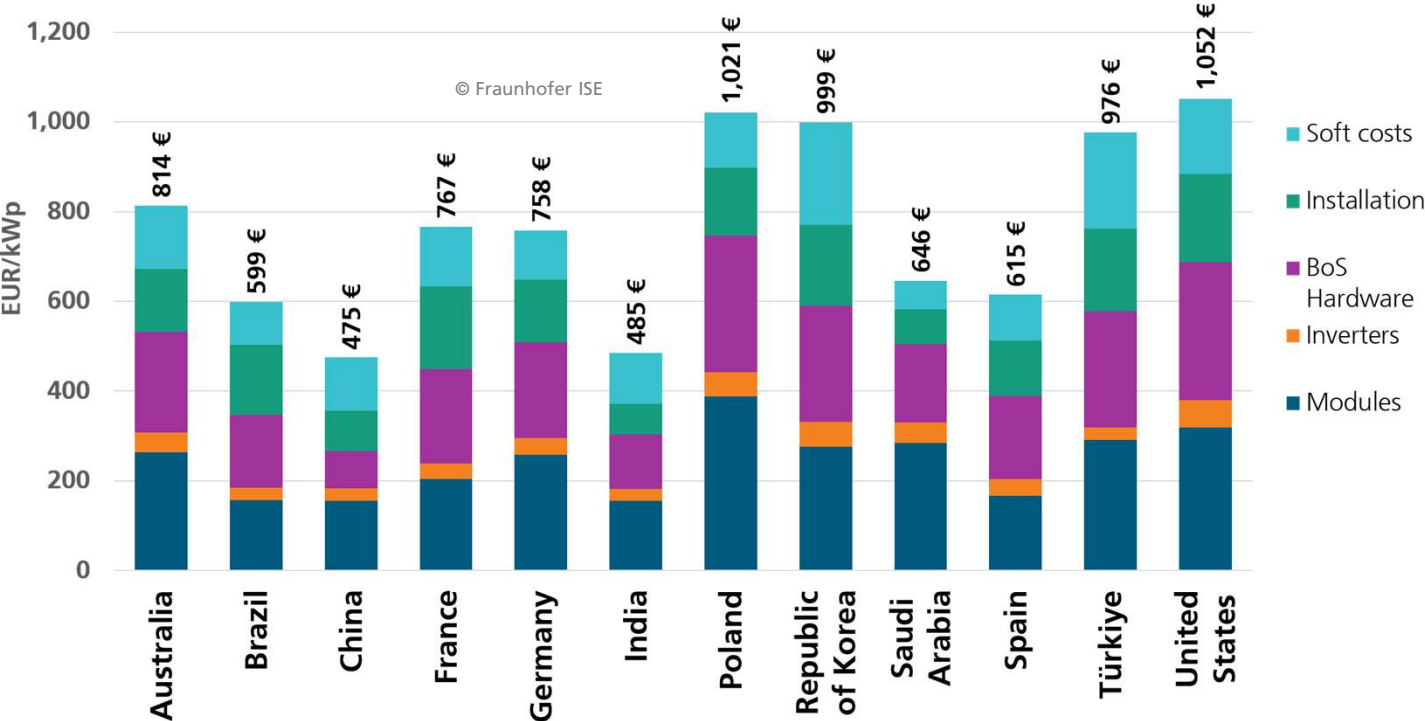


- The global weighted average total cost for large PV systems was 595 €/kWp in 2025.
- The 5th percentile is a value associated with the location within the data where 5% of the data is below this value. For 2025 the 5th percentile is 341 €/kWp.
- The 95th percentile is the value where 5% of the data has a higher value. For 2025, the 95th percentile is 1079 €/kWp.
- Over the past 15 years, total installed cost for utility-scale PV systems have fallen by an average of around 12% compared to the previous year.

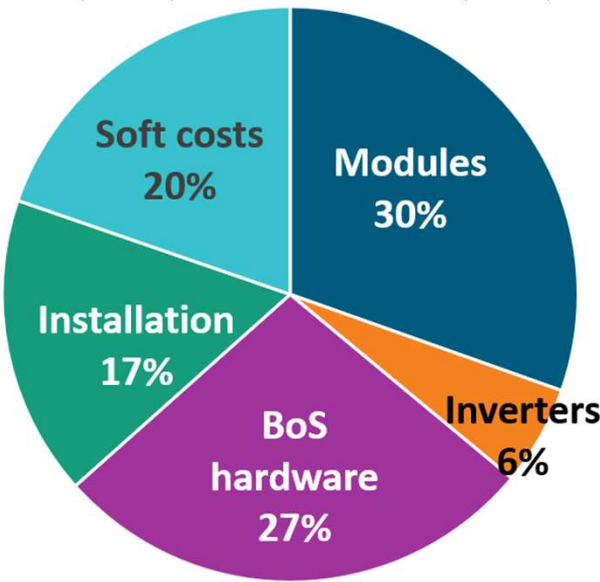
Data: IRENA (2026), Renewable Power Generation Costs in 2025, International Renewable Energy Agency, Abu Dhabi. Currency converted from USD to EUR. Date of data: July 2026
Currency converted from USD to EUR

Breakdown of Utility-Scale PV Total Installed Costs

By Country in 2025



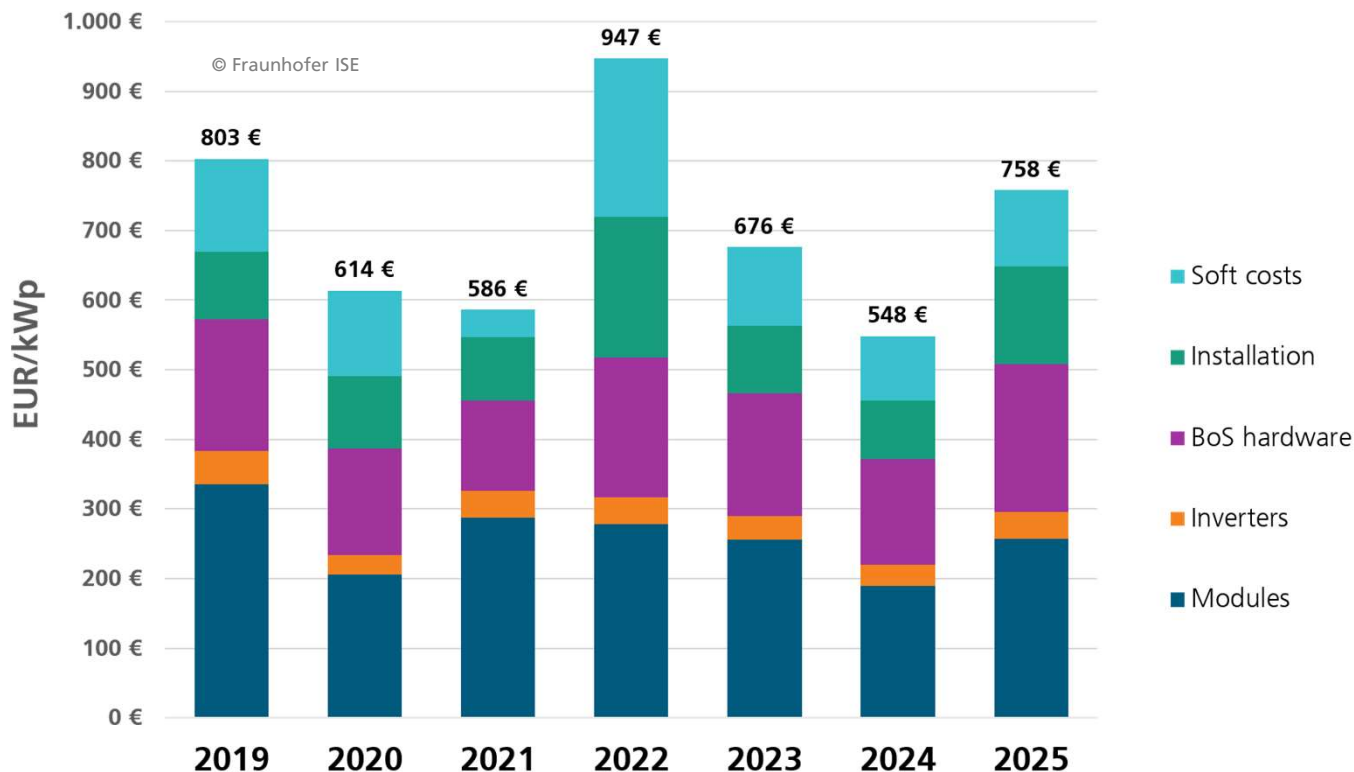
Breakdown of Cost Components
(average of available country data):



Data: IRENA (2026), Renewable Power Generation Costs in 2025, International Renewable Energy Agency, Abu Dhabi. Currency converted from USD to EUR. Date of data: July 2026

Breakdown of Total Installation Costs of Utility-Scale PV

Germany 2019 to 2025



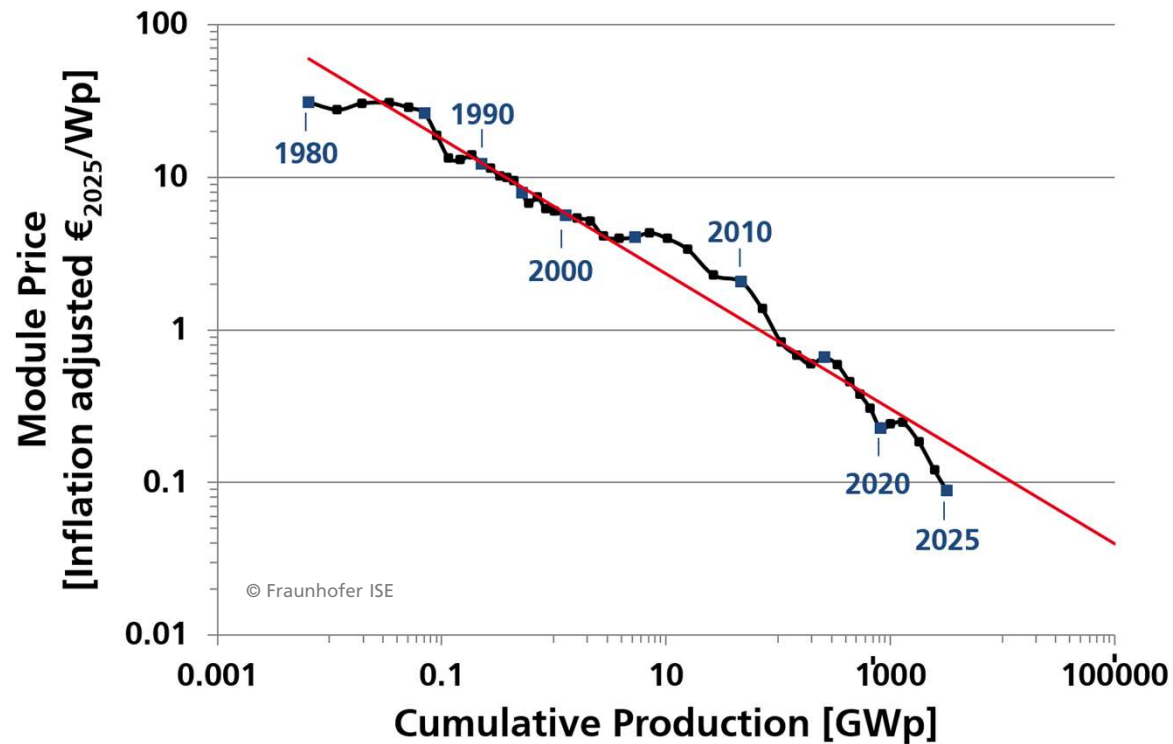
Supply shortages due to the coronavirus crisis led to price turbulence in 2022.

The rise in total installation costs in Germany in 2025 is mainly due to higher installation costs and soft costs such as financing, but also to an increase in hardware prices.

Data: IRENA (2026), Renewable Power Generation Costs in 2025, International Renewable Energy Agency, Abu Dhabi. Currency converted from USD to EUR. Date of data: July 2026

Price Experience Curve

Includes all Commercially Available PV Technologies

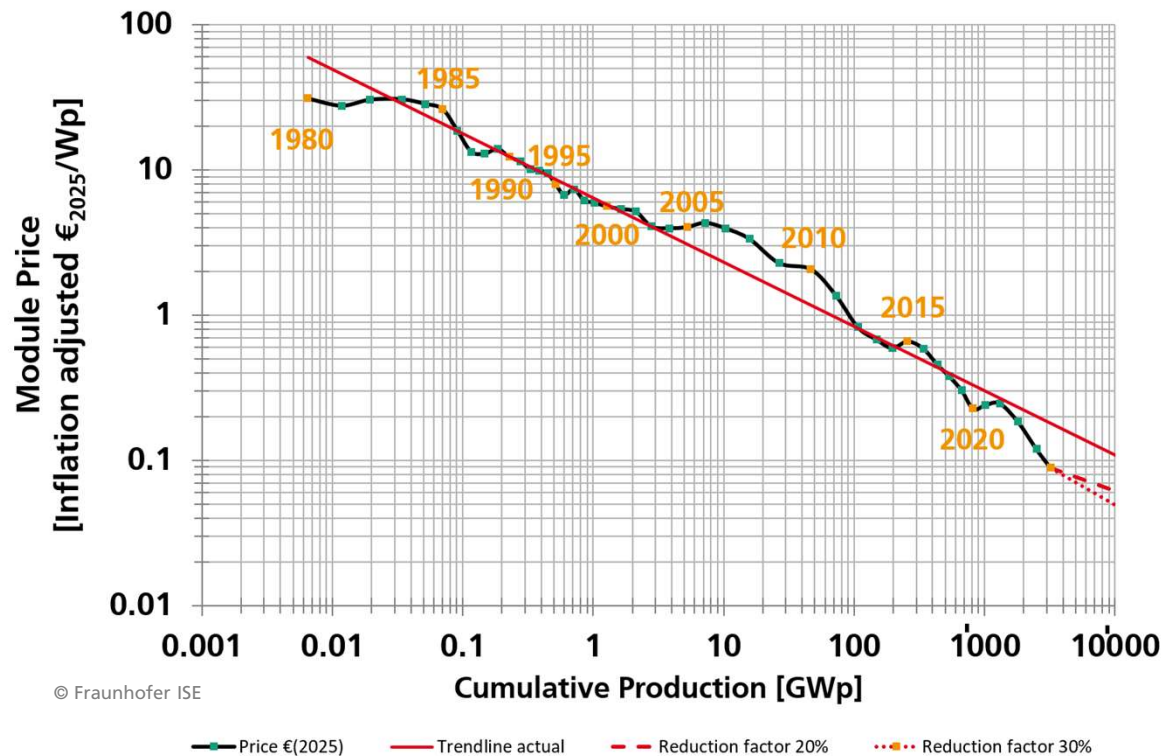


Learning Rate:
Each time the cumulative PV module production doubled, the module price dropped about 26.6% over the past 45 years.

Data: from 1980 to 2010 estimation from different sources: Strategies Unlimited, Navigant Consulting, EUPD, pvXchange; from 2011 until 2021: IHS Markit; since 2022 ITRPV/VDMA: ISE; Graph: PSE Projects GmbH 2026.

Price Experience Curve

Includes all Commercially Available PV Technologies



Learning Rate:
Each time the cumulative PV module production doubled, the module price dropped about 26.6% over the past 45 years.

Data: from 1980 to 2010 estimation from different sources: Strategies Unlimited, Navigant Consulting, EUPD, pvXchange; from 2011 until 2021: IHS Markit; since 2022 ITRPV/VDMA: ISE; Graph: PSE Projects GmbH 2026

Further Reading

Selected studies and analyses

[Fraunhofer-ISE Energy Charts](#)

[Study: Levelized Cost of Electricity - Renewable Energy Technologies](#)

[Recent facts about photovoltaics in Germany](#)

[Power Generation from Renewable Energy in Germany](#)

[What will the Energy Transformation Cost? Pathways for Transforming the German Energy System by 2050](#)

[Sustainable PV Manufacturing in Europe – An Initiative for a 10 GW Green Fab](#)

[Meta Study: Future Crosssectoral Decarbonization Target Systems in Comparison to Current Status of Technologies](#)

Abbreviations

Abbreviation	Explanation	Abbreviation	Explanation
AC	Alternating Current	HCPV	High Concentrator Photovoltaic
Al-BSF	Aluminum Back Surface Field	HJT (also HIT)	Heterojunction with Intrinsic Thin-Layer
BESS	Battery Energy Storage Systems	IBC	Interdigitated Back Contact (solar cells)
BIPV	Building Integrated PV	LCOE	Levelized Cost of Energy
BOS	Balance of System	LCPV	Low Concentrator Photovoltaic
CdTe	Cadmium-Telluride	MJ	Multi Junction
CI(G)S	Copper Indium (Gallium)Diselenide	MPP	Maximum Power Point
CPV	Concentrating Photovoltaic	n-type	Negatively doped wafer (with phosphorous)
c-Si	Crystalline Silicon	PERC	Passivated emitter and rear cell
CTM	Cell-to-Module	PR	Performance Ratio
Cz	Czochralski Method	p-type	Positively doped wafer (with boron or gallium)
DC	Direct current	PV	Photovoltaic
EEG	Renewable Energy Source Act (Erneuerbare-Energien-Gesetz, EEG)	RE	Renewable Energies
EI	The Energy Institute	ROI	Return on Investment
EPBT	Energy Payback Time	SI	Silicon
EROI	Energy Return of Invest	SIC	Silicon carbide
FZ	Floating Zone	TOPCon	Tunnel Oxide Passivated Contact
GaAs	Gallium Arsenide	VAT	Value Added Tax
GaN	Gallium nitride		

50

Acknowledgements

This work has been carried out with contributions from:

Name	Institution
Andreas Bett	ISE
Bruno Burger	ISE
Christoph Kost	ISE
Holger Neuhaus	ISE
Sebastian Nold	ISE
Simon Philipps	ISE
Ralf Preu	ISE
Christian Reichel	ISE
Jochen Rentsch	ISE
Gerhard Stryi-Hipp	ISE
Tobias Reuter	ISE
Harry Wirth	ISE
Werner Warmuth	PSE Projects GmbH

The information provided in this Photovoltaics Report is very concise by its nature . Its principal purpose is to provide a rough overview about the current solar PV market, the technologies and the environmental impact.

However, there are many more aspects. These and further details can be provided by Fraunhofer ISE upon request. Please contact us if you are interested in receiving a tailor-made offer.

simon.philipps@ise.fraunhofer.de
warmuth@pse-projects.de

Thank You
for Your Interest

Contact

simon.philipps@ise.fraunhofer.de

warmuth@pse-projects.de